Highlights

Proliferating Cell Collectives: A Comparison of Hard and Soft Collision Models

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- Hard and soft collision models for proliferating bacteria are systematically compared
- Both models reproduce macroscopic concentric ring patterns and microdomain formation
- Hard model enforces strict non-overlap; soft model shows unphysical cell overlap
- Hard model achieves up to 9.36× speedup via 30× larger step-sizes
- CFL-adaptive timestepping proves essential for efficient colony growth simulation

Proliferating Cell Collectives: A Comparison of Hard and Soft Collision Models

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Abstract

Dense ~~bacterial~~ microbial colonies self-organize into striking spatial patterns driven by mechanical feedback between growing cells. It is unclear whether rigid constraint-based (hard) or soft potential-based ~~approaches~~ collision models offer a better trade-off between physical accuracy and computational cost when simulating these dynamics at large scale.

~~This work systematically compares hard (constraint-based) and soft (potential-based)~~ We systematically compare hard and soft collision models for ~~simulating~~ proliferating cell collectives within a unified computational framework. Both reproduce the experimentally observed concentric ring patterns and microdomain formation~~in bacterial colonies, indicating~~, indicating that these macroscopic patterns are robust to the specific details of collision resolution.

At the ~~microscopic scale, the models differ significantly~~cell scale, however, the models diverge sharply. The hard model enforces strict non-overlap ~~, maintaining realistic packing fractions around 0.9 and producing accurate stress distributions and microdomain structures. The soft model allows substantialoverlap, with packing fractions exceeding 5 in colony centers, resulting in distorted microdomains and elongated cell bundles, limiting applicability for cell-scale analyses.~~ and maintains realistic packing throughout the colony, whereas the soft model permits substantial, unphysical overlap that distorts microdomain structure and cell-scale stresses, limiting its use for microscopic analyses. The hard model is also seen to be faster, remaining numerically stable at much larger step-sizes and scaling better in parallel.

~~Performance benchmarks show the hard model consistently outperforms the soft model, achieving up to 9.36 times fasterruntimes on 112 cores due to step-sizes roughly 30 times larger while maintaining numerical stability and better parallel scaling. Adaptive timestepping based on the Courant-Friedrichs-Lewy condition proves essential for both models to handle the rapidly changing configurations during colony growth.~~

~~Overall,~~ Within the regimes studied here – two-dimensional, mechanically-driven colonies of a single rod-shaped cell type – our results suggest that the hard model is ~~strongly preferredfor nearly all applications~~preferred, while the soft model suits ~~only~~ small-colony exploratory studies focused on macroscopic pattern formation.

1. Introduction

The collective behaviors of biological entities, from microbial colonies to multicellular tissues, exemplify how local interactions among individual organisms can generate complex, large-scale structures and dynamics. These systems often exhibit emergent spatio-temporal patterns that arise from the interplay of cell growth, division, mechanical interactions, and local environmental feedback.

An example is found in bacterial colonies, where cell arrangement reflects growth dynamics and internal stresses. Continuous cell proliferation within colonies generates significant mechanical forces that accumulate in the densely packed interior [37], leading to compressive stresses that suppress cell growth at the colony center. As a result, peripheral cells expand outward over time while internal cells remain densely packed and growth-limited. This outward expansion produces the characteristic concentric ring patterns observed in bacterial species such as *E. coli*, *Bacillus subtilis*, and *Proteus mirabilis*, as well as fungi like *Setosphaeria rostrata* and *Exserohilum turcicum* [38], as illustrated in Figure 1.

Computational modeling offers a powerful tool for dissecting the complex dynamics and emergent spatial patterns observed in those colonies. To reproduce such emergent structures and uncover their underlying mechanisms, models must accurately capture the dynamic interplay between cell growth and division, mechanical forces, and intercellular interactions.

Simulating dense, proliferating cell collectives, however, poses significant computational challenges: the large number of cells in realistic colonies, combined with the need to resolve collisions accurately across continuously changing configurations, creates a computational bottleneck.

Moreover, different collision-handling approaches offer distinct trade-offs in physical fidelity, numerical stability, and computational cost. Soft repulsive models are computationally efficient, but require small step-sizes; whereas hard

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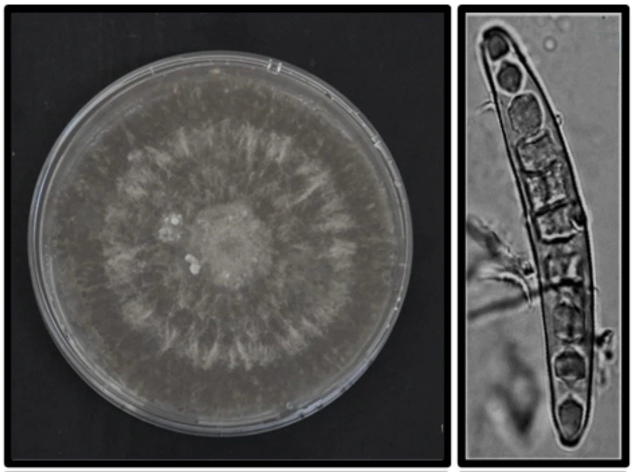


Figure 1: Morphology of the fungus *Exserohilum turcicum*. The left image shows a typical colony with concentric ring patterns, while the right image displays the elongated, rod-shaped morphology of individual fungal cells. Adapted from Bankole et al. [3] (*BMC Plant Biology*, 2023), licensed under CC BY 4.0 .

constraint-based models are computationally more expensive per timestep[1], but are more numerically stable allowing larger step-sizes. Whilst both approaches have been used in cell collective modeling, to the best of the authors' knowledge, no work has yet compared the two, particularly with a focus on computational efficiency. This work addresses this gap.

To this end, we provide a unified computational framework, written in C++ and utilizing PETSc and MPI, for the large-scale, efficient simulation of such cell colonies, which can use either hard or soft collision models. The simulator, mathematically, closely follows that of Weady et al.[35], who demonstrated the modeling of bacterial colony growth with a constraint-based approach. In addition, we implement a soft repulsive collision model, often used in other literature. To enable efficient large-scale parallelization, we propose a domain decomposition tailored to such radially growing colonies. To ensure fairness where both models require different step-sizes, we present an adaptive timestepping scheme that maintains stability in either model without requiring excessively small step-sizes. With this computational framework, we provide a systematic comparison of the two models, examining their respective strengths and limitations in simulating large systems of proliferating cells.

2. Related Work

2.1. Collision Modeling Paradigms

In computational biology, individual-based models (IBMs), also known as agent-based models, represent individual cells as discrete entities that grow, divide, and interact mechanically. Unlike continuum models, which treat populations as continuous fields, IBMs simulate single-cell behavior and local heterogeneity. A central computational challenge in such models is resolving cell-cell collisions efficiently. Two primary paradigms exist for handling collisions: soft (potential-based) collision models and hard (constraint-based) collision models.

Soft (Potential-Based) Models

Soft models manage cell interactions through repulsive forces defined by potentials such as Hertzian or Lennard-Jones functions. When cells overlap, these forces push them apart continuously. The primary advantage is computational efficiency: the calculations are local and easily parallelizable, enabling large-scale simulations. For instance, Warren et al. [34] demonstrated this by simulating millions of cells in three dimensions using a soft repulsion approach.

Beyond this, a number of studies using soft repulsion have yielded substantial biological insights into growing colonies. These also use Hertzian contact models and limit cell growth as nutrients deplete. Farrell et al., 2013, discussed how mechanical interactions shape the advancing colony front [14], and Farrell et al., 2017, demonstrated that mechanical properties of cells influence the probability that a cell with a faster-growth mutation will 'surf' the advancing colony front as opposed to forming only a small cluster inside the bulk of the colony [13]. Gosh & Levine explored the impact of cell death upon the cell colony dynamics, considering both cell death through limited nutrients and programmed cell death [16]. Ghosh et al. explore the mechanical impact of extracellular polymers modeled as small spherical particles secreted by the cells, which can phase-separate cells into clusters separated by polymers [17], and Bera et al. expand this to explore cell motility (self-propelled cells), showing that higher motility results in sparser colonies without phase separation [4]. It is not only purely repulsive models which have been used in literature: Bera et al. introduce an attractive Lennard-Jones force to mimic the sticky nature of such extracellular polymers [5].

However, soft models suffer from two key limitations. First, the stiffness of repulsive potentials creates numerical instability, requiring very small step-sizes to maintain accuracy and prevent divergence. The code supplied with Warren et al. [34] provides an input file with a step-size of 5×10^{-5}. None of the other papers mentioned above provides step-size information, but to maintain stability, we presume similar or smaller step-sizes are used. Second, the finite range of repulsive forces means cells behave as elastically deformable objects rather than rigid bodies, effectively reducing their geometric diameter below the intended size [41]. This deformation accumulates

[1] Regarding terminology: In this work, we refer to an iteration of a numerical iterator such as Explicit Euler, as a *timestep*. We refer to the change in time between one timestep and the next as the *step-size*.

and can distort simulation results, particularly in dense regions.

Hard (Constraint-Based) Models
Hard models enforce strict non-overlapping constraints, treating cells as geometrically rigid and impenetrable. Constraint-based methods are widely used in physics simulations [15, 23, 24, 32, 40], and have also been adapted for biological systems [28, 35, 41].

The main drawback is computational cost and added complexity. Enforcing global non-overlap constraints requires iterative solvers operating on large coupled systems of equations, which creates a significant bottleneck. Modern implementations, however, can efficiently reduce the numerical stiffness associated with soft models, enabling much larger step-sizes while maintaining geometric fidelity [41]. Nonetheless, the fundamental trade-off remains: higher cost per timestep due to constraint solving, but fewer timesteps required overall.

In addition to such IBMs, continuum models are widely used in this field. Giverso et al. study pattern formation and branching at the interface between the colony and the substrate [19], and Weady et al. [35] derive a continuum model that matches their IBM. For a broader review of continuum models, we refer the reader to Porter et al., which provides an extensive review of the topic, focusing more heavily on continuum models than on IBMs [26].

2.2. The Benchmarking Gap

While soft and hard collision models entail distinct computational trade-offs, a direct, systematic comparison of their performance remains unexplored. This trade-off becomes particularly critical when simulating large populations of independent agents, where the choice of collision model directly affects runtime, scalability, and the accuracy of the simulation.

Existing biological studies often validate their approaches against experimental data, such as microscopy images, growth curves, and morphological features [6, 17, 21, 22, 27, 28, 33, 34, 35, 42, 43], with a primary emphasis on biological fidelity. Comparatively fewer works examine computational aspects such as runtime, scalability, and numerical stability across varying colony sizes, leaving limited guidance on selecting an appropriate modeling paradigm for a given problem scale.

This work aims to address this gap by implementing both soft and hard collision schemes within a shared computational framework, enabling direct performance comparisons. We systematically benchmark the two approaches across a range of colony sizes, evaluating their respective strengths and limitations in both biological fidelity and computational efficiency.

3. Cell Mechanics

Unless otherwise noted, the mechanics model in this section follows Weady et al. [35, 36].

Accurately modeling mechanical interactions between cells is essential for simulating colony growth and emergent spatial patterns. From a computational perspective, approximating bacteria and fungi as rigid spherocylinders (rods with hemispherical end caps) offers a balance between biological realism and computational efficiency. This representation captures the elongated morphology of these organisms while preserving the fundamental mechanics of cell-cell interactions and colony expansion. Although real cells exhibit some deformability, the rigid spherocylinder approximation successfully captures essential collective dynamics at significantly reduced computational cost and is well validated experimentally [17, 21, 22, 27, 28]. The spherocylinder geometry closely matches common organisms such as *E. coli* and *Setosphaeria rostrata* (see Figure 1), and has become standard in computational models of microbial collectives [6, 17, 34, 35, 43]. Although we refer to bacterial colonies throughout for consistency, the model targets rod-shaped microbial collectives in general, and applies equally to the rod-shaped fungi noted above.

Each spherocylinder is characterized by a time-varying length $\ell_i(t)$ that increases due to growth prior to division, and a fixed diameter d that remains constant across the population. This geometry enables efficient contact detection algorithms [12] and naturally accommodates rotational dynamics essential for capturing realistic colony mechanics.

Although bacteria are fundamentally three-dimensional objects, the simulations in this work are restricted to a two-dimensional plane, consistent with early colony growth on a flat substrate. All quantities, including forces, torques, and orientations, are still represented as three-dimensional vectors; however, the third spatial dimension is not utilized ~~. This effectively reduces the dynamics to a planar system while retaining a 3D vector formulation, and the extension to fully~~ in this work. Nevertheless, the framework itself is natively three-dimensional, with all quantities, like force and torque, fully represented, and could be used to conduct three-dimensional simulations~~is straightforward.~~ : contact detection operates directly on 3D line segments [12, 41], and full spatial dynamics are recovered simply by enabling the out-of-plane force and noise components. To demonstrate this, our supplementary materials include a simulation of a colony growing within a bounded cube, in which additional virtual wall constraints physically confine the cells.

At bacterial length scales, the Reynolds number is extremely small (Re $\ll$ 1), so viscous forces dominate over inertia. In this highly viscous, *sticky* environment, cells stop almost immediately once forces cease [11, 28]. As a result, cell motion is governed by the overdamped Langevin equation:

$$\mathbf{u}_i = \frac{1}{\zeta \ell_i} \sum_{j \neq i} \mathbf{F}_{ij}, \quad \boldsymbol{\omega}_i = \frac{12}{\zeta \ell_i^3} \sum_{j \neq i} \mathbf{r}_{ij} \times \mathbf{F}_{ij} \tag{1}$$

Here, $\mathbf{u}_i$ and $\boldsymbol{\omega}_i$ are the translational and angular velocities of cell i, $\mathbf{F}_{ij}$ is the force exerted on cell i by cell j, ζ is the drag coefficient, and $\mathbf{r}_{ij}$ is the lever arm from the cell center to the contact point. Figure 2 illustrates the model and the forces and torques during collision.

3.1. Stress-Dependent Growth

Bacterial cells grow along their longitudinal axis according to the following model, adapted from [36]:

$$\dot{\ell}_i = \frac{\ell_i}{\tau} e^{-\lambda' \sigma_i} \tag{2}$$

where τ is the characteristic division time under zero stress, σ_i is the compressive stress along cell i's main axis, and λ' is a stress sensitivity parameter. As the compressive stress increases, the effective growth rate $\dot{\ell}_i$ decreases toward zero, leading to slower elongation. When $\sigma_i = 0$, the cell grows exponentially with a rate determined solely by τ.

The stress on cell i is computed by projecting contact forces onto the cell's longitudinal axis $\hat{\mathbf{t}}_i$:

$$\sigma_i = \sum_{j \neq i} \frac{1}{2} \left| \hat{\mathbf{t}}_i \cdot \mathbf{F}_{ij} \right| \tag{3}$$

Following [36], we nondimensionalize lengths by ℓ_0 (the reference cell length at birth), time by τ, and stress by $\tau/(\zeta \ell_0^2)$. This leaves the dimensionless growth sensitivity $\lambda = (\tau/\zeta \ell_0^2)\lambda'$ as the only free parameter, and we fix $\ell_0 = 1$, $d = 0.5$, $\tau = 1$, $\zeta = 1$ throughout.

Cells divide at a critical length $\ell_{\text{crit}} = 2\ell_0$, producing two daughters with lengths uniformly randomly sampled from $[0.98\ell_0, 1.02\ell_0]$ to avoid synchronized divisions [43].

3.2. Rotational Diffusion

To account for Brownian motion and other sources of random fluctuations, we apply a small random rotational perturbation to each cell at every timestep. This is particularly important in the early colony phase, where few mechanical interactions exist to break the orientational symmetry: without noise, cells that happen to be aligned would remain so indefinitely, producing artificially ordered initial configurations that do not reflect biological reality. The magnitude of the perturbation was determined empirically to produce realistic spreading behavior in very small colonies of only a few cells, while remaining small enough not to artificially disorder larger, mechanically-driven colonies.

4. Unified Computational Framework

Both models share an identical colony representation, growth law, stress computation, and timestepping framework. The only difference lies in the computation of contact force magnitudes. In both formulations, forces are aligned with the contact normal. In the hard-contact model, force magnitudes are obtained via Lagrange multipliers enforcing a non-overlap constraint, whereas in the soft-contact model they arise from a local repulsive potential. All subsequent mechanical responses are identical in form, such that any observed differences in behavior or computational performance can be attributed solely to the contact formulation. This controlled setup enables a direct, systematic comparison of hard and soft interaction models, which have previously been studied in isolation.

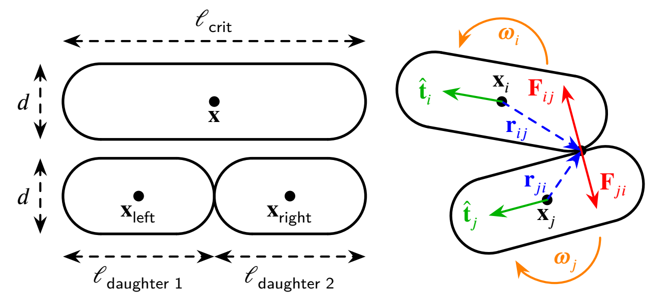


Figure 2: Spherocylinder cell model. Cells are modeled as 3D spherocylinders; the diagram shows a 2D schematic view. Contact forces $\mathbf{F}_{ij}$ (red) and lever arms $\mathbf{r}_{ij}$ (dashed blue) generate torques causing angular motion $\boldsymbol{\omega}$ (orange), and also produce translational displacements of the cells. The longitudinal axis $\hat{\mathbf{t}}_i$ (green) defines the direction along which stress is projected. Cell division occurs when a cell reaches critical length ℓ_{crit}.

4.1. Colony Representation

Following Weady et al. [36], the bacterial colony is represented by a global state vector $\mathcal{C} = [\dots, \mathbf{x}_n^\top, \mathbf{q}_n^\top, \dots]^\top \in \mathbb{R}^{7N}$, where N is the number of cells. Each cell is represented by a 7-dimensional sub-vector consisting of a position vector $\mathbf{x}_n \in \mathbb{R}^3$ and a unit quaternion $\mathbf{q}_n \in \mathbb{R}^4$ representing its orientation. The lengths of all cells are stored separately in a vector $\boldsymbol{\ell} = [\dots, \ell_n, \dots]^\top \in \mathbb{R}^N$. Quaternions are used instead of Euler angles or rotation matrices because they avoid singularities and provide stable numerical integration over long simulations.

4.2. Colony Dynamics

~~The translational and angular velocities of all cells in the colony from are determined simultaneously by a force-velocity relationship $\mathcal{U} = \mathcal{M}^k \mathcal{F}$. Here,~~ We define $\mathcal{F} \in \mathbb{R}^{6N}$ ~~is~~ as the generalized force vector for the entire colony, containing both force and torque components for each cell~~. Similarly,~~, and $\mathcal{U} \in \mathbb{R}^{6N}$ ~~is~~ as the generalized velocity vector containing the resulting translational and angular velocities~~. The vectors are defined~~, as:

$$\begin{aligned} \mathcal{F} &= [\, \mathbf{f}_1, \boldsymbol{\tau}_1, \dots, \mathbf{f}_N, \boldsymbol{\tau}_N]^\top \in \mathbb{R}^{6N}, \\ \mathcal{U} &= [\mathbf{u}_1, \boldsymbol{\omega}_1, \dots, \mathbf{u}_N, \boldsymbol{\omega}_N]^\top \in \mathbb{R}^{6N} \end{aligned} \tag{4}$$

where, $\mathbf{f}_i = \sum_j \mathbf{F}_{ij}$ is the total force on cell i due to all other cells j in contact with it, and $\boldsymbol{\tau}_i = \sum_j \mathbf{r}_{ij} \times \mathbf{F}_{ij}$ is the total torque. ~~The~~

The translational and angular velocities of all cells in the colony from Equation 1 are then determined simultaneously by a force-velocity relationship $\mathcal{U} = \mathcal{M}^k \mathcal{F}$, where $\mathcal{M}^k$

is the diagonal mobility matrix $\cancel{\mathcal{M}^k}$ at timestep k~~encodes~~, encoding the drag coefficients for all cells based on their current lengths ℓ_i^k:

$$\mathcal{M}^k = \operatorname{diag}\left(\left\{ \frac{1}{\zeta \ell_i^k} \mathbf{I}_3,\ \frac{12}{\zeta (\ell_i^k)^3} \mathbf{I}_3 \right\}_{i=1}^{N} \right) \in \mathbb{R}^{6N \times 6N} \tag{5}$$

where $\mathbf{I}_3$ is the 3×3 identity matrix. The resulting matrix-vector product thus implements the overdamped Langevin equation from Equation 1 for all cells simultaneously.

4.3. State Integration

The orientation quaternion $\mathbf{q}_n^k$ for each cell evolves according to the kinematic equation $\dot{\mathbf{q}}_n^k = \frac{1}{2}\boldsymbol{\omega}_n \otimes \mathbf{q}_n^k$, where $\otimes$ denotes quaternion multiplication. To integrate translational and rotational dynamics consistently, we introduce a mapping $\mathcal{G}^k \in \mathbb{R}^{7N \times 6N}$, converting Cartesian translational and angular velocities from $\mathbb{R}^{6N}$ into corresponding changes in positions and quaternions in $\mathbb{R}^{7N}$ according to the quaternion kinematics (see [32, 36, 39]). Using $\mathcal{G}^k$, the colony's state and the length vector are updated via explicit Euler integration with step-size Δt.

$$\begin{aligned} \mathcal{C}^{k+1} &= \mathcal{C}^k + \Delta t\, \mathcal{G}^k \mathcal{U} = \mathcal{C}^k + \Delta t\, \mathcal{G}^k \mathcal{M}^k \mathcal{F} \\ \boldsymbol{\ell}^{k+1} &= \boldsymbol{\ell}^k + \Delta t\, \dot{\boldsymbol{\ell}} \end{aligned} \tag{6}$$

With the shared framework established, the two models differ only in how the inter-cell forces $\mathbf{F}_{ij}$ (and hence $\mathcal{F}$) are computed, as described in the following subsections.

4.4. Soft Collision Model

The soft collision model treats cell-cell interactions through continuous, potential-based repulsive forces. When cells overlap, these forces smoothly increase, preventing significant inter-cell penetration while allowing elastic deformation. This approach is well-suited for simulating the soft, deformable nature of real biological cells and has been widely used in prior work [6, 17, 21, 22, 27, 33, 34, 42, 43].

We implement a Hertzian contact model, which describes elastic deformation between curved surfaces in contact. The repulsive elastic force exerted on cell i by cell j is:

$$\mathbf{F}_{ij}^{\text{elastic}} = k_{cc} \sqrt{d}\, \delta^{3/2}\, \hat{\mathbf{n}} \tag{7}$$

where δ is the overlap distance, d is the cell diameter, k_{cc} is an elastic constant, and $\hat{\mathbf{n}}$ is the unit normal vector at the contact point. The $\delta^{3/2}$ dependence is characteristic of Hertzian theory for elastic contacts and produces a nonlinear, smooth increase in force as overlap grows.

4.4.1. Force Assembly

The total force and torque on cell n are computed by summing contributions from all pairwise elastic interactions and can be used to assemble the global force vector $\mathcal{F}$ and compute the cell growth rates $\dot{\boldsymbol{\ell}}$ via Equation 2:

$$\mathbf{f}_i = \sum_{j \neq i} \mathbf{F}_{ij}^{\text{elastic}}, \qquad \boldsymbol{\tau}_i = \sum_{j \neq i} \mathbf{r}_{ij} \times \mathbf{F}_{ij}^{\text{elastic}} \tag{8}$$

4.4.2. Model Parameters and Numerical Stability

The elastic constant is defined as $k_{cc} = \frac{Y}{\zeta}$, where Y is the cell's Young's modulus and ζ is the fluid drag coefficient. Using typical bacterial parameters ($Y \approx 4$ MPa and $\zeta \approx 200$ Pa·h) [6, 43], we set $k_{cc} = 20000\ \text{h}^{-1}$. This value is sufficiently large to prevent significant overlap between cells. However, the steep, nonlinear nature of the Hertzian force law creates numerical stiffness: cells can undergo rapid acceleration when overlap occurs, potentially causing large position changes within a single timestep. To maintain stability, an extremely small step-size ($\Delta t \sim 10^{-5}$) is required [6, 21, 43].

This step-size limit restriction is a well-known limitation of soft collision models and creates a significant computational bottleneck for large colonies.

4.5. Hard Collision Model

The hard collision model adapted from Weady et al. [36] enforces strict non-overlapping constraints between cells using a constraint-based method. For each nearby cell pair, a constraint α is defined based on the closest points on their surfaces, $\mathbf{y}_n$ and $\mathbf{y}_m$. Unlike the soft model, which uses repulsive forces, the hard model determines contact forces via unknown scalar Lagrange multipliers γ_α, which represent the magnitude of the impulse needed to prevent penetration.

The force on cell n from constraint α is:

$$\mathbf{F}_{n\alpha}^{\text{hard}} = \hat{\mathbf{n}}_\alpha \gamma_\alpha \tag{9}$$

where $\hat{\mathbf{n}}_\alpha$ is the normal vector at the contact point. The multipliers $\boldsymbol{\gamma}$ are solved fresh each timestep and carry no history. The total generalized force vector $\mathcal{F}$ for the colony depends linearly on the multipliers $\boldsymbol{\gamma} = [\dots, \gamma_\alpha, \dots]^\top \in \mathbb{R}^C$, where C is the number of constraints, and can be conveniently expressed in matrix form:

$$\mathcal{F}(\boldsymbol{\gamma}) = \mathcal{D}\boldsymbol{\gamma} \tag{10}$$

where $\mathcal{D} \in \mathbb{R}^{6N \times C}$ is a sparse matrix consisting of contact normals and lever arms for all constraints. This matrix-vector product mimics the force assembly in Equation 8, but uses the unknown multipliers $\boldsymbol{\gamma}$ instead of explicit force calculations. Similarly, the stress vector $\boldsymbol{\sigma} = [\dots, \sigma_n, \dots]^\top \in \mathbb{R}^N$ can also be expressed linearly in terms of $\boldsymbol{\gamma}$:

$$\boldsymbol{\sigma}(\boldsymbol{\gamma}) = \mathcal{L}\boldsymbol{\gamma} \tag{11}$$

where $\mathcal{L} \in \mathbb{R}^{N \times C}$ encodes stress contributions from each constraint, such that the matrix-vector product mimics the stress calculation in Equation 3. This also means that the growth rates $\dot{\boldsymbol{\ell}}$ depend (nonlinearly) on $\boldsymbol{\gamma}$ via Equation 2.

Together, these linear maps connect the single unknown $\boldsymbol{\gamma}$ to every physical quantity in the update. Figure 3 summarizes how the force map $\boldsymbol{\mathcal{D}}$, the stress map $\boldsymbol{\mathcal{L}}$, and the integration map $\boldsymbol{\mathcal{G}}$ propagate the multipliers into forces, stresses, and the updated colony state, and how their transposes feed the resulting motion and growth back into the separation distances.

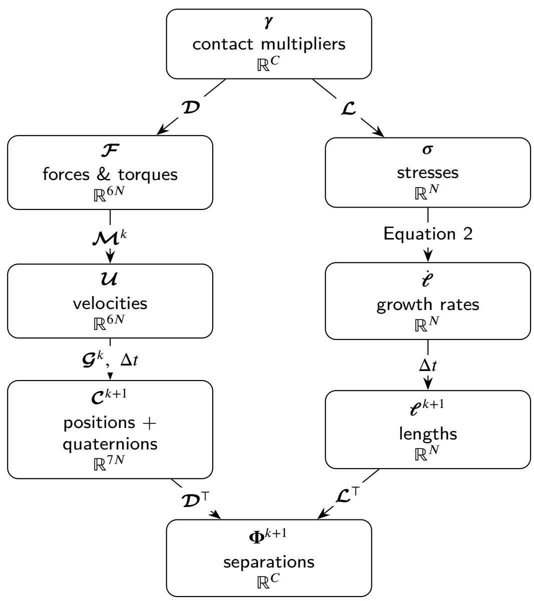


Figure 3: Data flow of the hard collision model. The single unknown—the vector of contact multipliers $\boldsymbol{\gamma}$ (one per constraint)—generates the generalized forces $\boldsymbol{\mathcal{F}}$ via $\boldsymbol{\mathcal{D}}$ and the per-cell stresses $\boldsymbol{\sigma}$ via $\boldsymbol{\mathcal{L}}$. The mobility $\boldsymbol{\mathcal{M}}^k$ maps forces to velocities $\boldsymbol{\mathcal{U}}$, while the stress-dependent growth law (Equation 2) sets the elongation rates $\dot{\boldsymbol{\ell}}$. The integration map $\boldsymbol{\mathcal{G}}^k$ advances velocities into position and quaternion updates. The updated configuration $(\boldsymbol{\mathcal{C}}^{k+1}, \boldsymbol{\ell}^{k+1})$ fixes the new signed separations $\boldsymbol{\Phi}^{k+1}$, whose linearization $\boldsymbol{\Phi}^{k+1} = \boldsymbol{\Phi}^k + \Delta t(\boldsymbol{\mathcal{D}}^\top \boldsymbol{\mathcal{U}} - \boldsymbol{\mathcal{L}}^\top \dot{\boldsymbol{\ell}})$ (Equation 14) closes the complementarity problem for $\boldsymbol{\gamma}$. The transposes $\boldsymbol{\mathcal{D}}^\top$ and $\boldsymbol{\mathcal{L}}^\top$ thus act as the Jacobians mapping motion and growth back to changes in separation.

4.5.1. Constraint Conditions

To ensure that solving for the multipliers $\boldsymbol{\gamma}$ leads to physically meaningful results, three key conditions must be satisfied:

1. **Repulsive Forces:** $\boldsymbol{\gamma} \geq \mathbf{0}$. This ensures that the forces are repulsive, preventing cells from attracting each other.
2. **No-Overlap:** $\boldsymbol{\Phi}^{k+1} \geq \mathbf{0}$ with $\boldsymbol{\Phi}^{k+1} = [\dots, \Phi_\alpha^{k+1}, \dots]^\top$. Here, $\boldsymbol{\Phi}^{k+1} \in \mathbb{R}^C$ is the vector of signed separation distances between the pairs of cells associated with each constraint α at timestep $k+1$. Signed separation distances between cells are computed using the robust geometric algorithm described by [12, 41]. A positive value Φ_α^{k+1} indicates that the cells in the next iteration are separated, while a negative value indicates overlap. This condition enforces that no overlaps occur after the update.
3. **Complementarity Condition:** $\boldsymbol{\gamma}^\top \boldsymbol{\Phi}^{k+1} = \mathbf{0}$ (often written as $\boldsymbol{\gamma} \perp \boldsymbol{\Phi}^{k+1}$). If cells are separated ($\Phi_\alpha > 0$), no force acts ($\gamma_\alpha = 0$); if a force acts ($\gamma_\alpha > 0$), cells must be touching ($\Phi_\alpha = 0$).

These conditions are often abbreviated as:

$$\mathbf{0} \leq \boldsymbol{\gamma} \perp \boldsymbol{\Phi}^{k+1} \geq \mathbf{0}. \tag{12}$$

The colony update rule and additional conditions combine to form a system where the future state $\boldsymbol{\mathcal{C}}^{k+1}$ and $\boldsymbol{\ell}^{k+1}$, and thus $\boldsymbol{\Phi}^{k+1}$ depend on $\boldsymbol{\gamma}$:

$$\begin{aligned} \boldsymbol{\mathcal{C}}^{k+1} &= \boldsymbol{\mathcal{C}}^k + \Delta t \boldsymbol{\mathcal{G}}^k \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{F}}(\boldsymbol{\gamma}) \\ \boldsymbol{\ell}^{k+1} &= \boldsymbol{\ell}^k + \Delta t\, \dot{\boldsymbol{\ell}}(\boldsymbol{\gamma}) \\ \text{s.t.} \quad & \mathbf{0} \leq \boldsymbol{\gamma} \perp \boldsymbol{\Phi}^{k+1}(\boldsymbol{\gamma}) \geq \mathbf{0}. \end{aligned} \tag{13}$$

4.5.2. Linearization of the Constraint Condition

The central challenge is that $\boldsymbol{\Phi}^{k+1}(\boldsymbol{\gamma})$ is a nonlinear function of the updated state $\boldsymbol{\mathcal{C}}^{k+1}$ and the updated lengths $\boldsymbol{\ell}^{k+1}$, and is non-trivial to compute directly. Weady et al. [36] and Yan et al. [40] propose a linearization approach to approximate $\boldsymbol{\Phi}^{k+1}(\boldsymbol{\gamma})$ using a Taylor expansion around the current state, $\boldsymbol{\Phi}^k(\boldsymbol{\gamma})$:
The change in $\boldsymbol{\Phi}$ in our model arises from two main effects:

1. **Cell Motion:** All cell distances change due to the motion driven by collision forces. This is captured by the term $\dot{\boldsymbol{\Phi}}^k{}_{\text{motion}} = \frac{d\boldsymbol{\Phi}^k}{d\boldsymbol{\mathcal{C}}} \frac{d\boldsymbol{\mathcal{C}}}{dt} = (\nabla_{\boldsymbol{\mathcal{C}}} \boldsymbol{\Phi}^k) \dot{\boldsymbol{\mathcal{C}}}$.
2. **Cell Growth:** Additionally, cell growth affects separation according to $\dot{\boldsymbol{\Phi}}^k{}_{\text{growth}} = \frac{d\boldsymbol{\Phi}^k}{d\boldsymbol{\ell}} \frac{d\boldsymbol{\ell}}{dt} = -(\nabla_{\boldsymbol{\ell}} \boldsymbol{\Phi}^k) \dot{\boldsymbol{\ell}}$. The negative sign arises because as cells elongate, they reduce the separation distance between them.

Combining these two effects leads to the following approximation for $\boldsymbol{\Phi}^{k+1}$:

$$\begin{aligned} \boldsymbol{\Phi}^{k+1}(\boldsymbol{\gamma}) &\approx \boldsymbol{\Phi}^k + \Delta t \left(\dot{\boldsymbol{\Phi}}_{\text{motion}} + \dot{\boldsymbol{\Phi}}_{\text{growth}} \right) \\ &= \boldsymbol{\Phi}^k + \Delta t \left((\nabla_{\boldsymbol{\mathcal{C}}} \boldsymbol{\Phi}^k) \dot{\boldsymbol{\mathcal{C}}}^k - (\nabla_{\boldsymbol{\ell}} \boldsymbol{\Phi}^k) \dot{\boldsymbol{\ell}} \right) \\ &= \boldsymbol{\Phi}^k + \Delta t \left((\nabla_{\boldsymbol{\mathcal{C}}} \boldsymbol{\Phi}^k) \boldsymbol{\mathcal{G}}^k \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{F}}(\boldsymbol{\gamma}) - (\nabla_{\boldsymbol{\ell}} \boldsymbol{\Phi}^k) \dot{\boldsymbol{\ell}} \right) \\ &\overset{\dagger}{=} \boldsymbol{\Phi}^k + \Delta t \left(\boldsymbol{\mathcal{D}}^\top \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{F}}(\boldsymbol{\gamma}) - (\nabla_{\boldsymbol{\ell}} \boldsymbol{\Phi}^k) \dot{\boldsymbol{\ell}} \right) \\ &\overset{\ddagger}{=} \boldsymbol{\Phi}^k + \Delta t \left(\boldsymbol{\mathcal{D}}^\top \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{D}} \boldsymbol{\gamma} - \boldsymbol{\mathcal{L}}^\top \left(\frac{\boldsymbol{\ell}}{\tau} \odot e^{-\lambda \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}} \right) \right) \end{aligned} \tag{14}$$

where $\odot$ denotes the element-wise product.

Here (†), we use the previously introduced $\boldsymbol{\mathcal{D}}$, noting that $\boldsymbol{\mathcal{D}}^\top = (\nabla_{\boldsymbol{\mathcal{C}}} \boldsymbol{\Phi}^k) \boldsymbol{\mathcal{G}}^k$. The transpose $\boldsymbol{\mathcal{D}}^\top$ can thus be interpreted as a Jacobian mapping velocities in $\mathbb{R}^{6N}$ first to changes in the colony state (via $\boldsymbol{\mathcal{G}}^k$) and then to changes in the signed distance functions $\boldsymbol{\Phi}$ (see [32, 36, 40] for details).

Similarly, in (‡), the previously defined $\boldsymbol{\mathcal{L}}$, which encodes stress contributions from each constraint, has a transpose $\boldsymbol{\mathcal{L}}^\top$ that maps stress-induced changes in cell lengths $\boldsymbol{\ell}$ to variations in $\boldsymbol{\Phi}$ (see [36] for details).

4.5.3. Nonlinear Complementarity Problem

Substituting the approximation of $\mathbf{\Phi}^{k+1}(\boldsymbol{\gamma})$ from Equation 14 into the constraint conditions of Equation 13 yields a nonlinear complementarity problem (NCP) for $\boldsymbol{\gamma}$ which no longer depends on the unknown future states $\boldsymbol{\mathcal{C}}^{k+1}(\boldsymbol{\gamma})$ and $\boldsymbol{\ell}^{k+1}(\boldsymbol{\gamma})$:

$$\boxed{\begin{array}{l}\text{Find } \boldsymbol{\gamma} \text{ such that} \\ \mathbf{0} \leq \boldsymbol{\gamma} \perp \mathbf{\Phi}^k + \Delta t \left(\boldsymbol{\mathcal{D}}^\top \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{D}} \boldsymbol{\gamma} - \boldsymbol{\mathcal{L}}^\top \left(\frac{\boldsymbol{\ell}}{\tau} \odot e^{-\lambda \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}} \right) \right) \geq \mathbf{0}.\end{array}} \tag{15}$$

4.5.4. Energy Minimization Solution

To solve this NCP efficiently, we reframe it as an optimization problem using a special energy function $E(\boldsymbol{\gamma})$ whose minimization under non-negativity constraints resolves the contact forces [29, 40]. Physically, $E(\boldsymbol{\gamma})$ represents the work done by the contact forces and the system's internal energy:

$$\begin{aligned} \min_{\boldsymbol{\gamma} \geq \mathbf{0}} E(\boldsymbol{\gamma}) = \min_{\boldsymbol{\gamma} \geq \mathbf{0}} \Big[& \boldsymbol{\gamma}^\top \mathbf{\Phi}^k + \frac{\Delta t}{2} \boldsymbol{\gamma}^\top \boldsymbol{\mathcal{D}}^\top \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{D}} \boldsymbol{\gamma} \\ & + \mathbf{1}^\top \frac{\Delta t}{\lambda} \left(\frac{\boldsymbol{\ell}}{\tau} \odot e^{-\lambda \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}} \right) \Big] \end{aligned} \tag{16}$$

By construction, the gradient of this energy equals the linearized separation distances $\mathbf{\Phi}^{k+1}(\boldsymbol{\gamma})$ from Equation 14:

$$\begin{aligned} \nabla_{\boldsymbol{\gamma}} E &= \mathbf{\Phi}^k + \Delta t \left(\boldsymbol{\mathcal{D}}^\top \boldsymbol{\mathcal{M}}^k \boldsymbol{\mathcal{D}} \boldsymbol{\gamma} - \boldsymbol{\mathcal{L}}^\top \left(\frac{\boldsymbol{\ell}}{\tau} \odot e^{-\lambda \boldsymbol{\mathcal{L}} \boldsymbol{\gamma}} \right) \right) \\ &\approx \mathbf{\Phi}^{k+1}(\boldsymbol{\gamma}) \end{aligned} \tag{17}$$

Because $E(\boldsymbol{\gamma})$ is convex, the constrained optimization problem has a unique global minimizer over the feasible region $\boldsymbol{\gamma} \geq \mathbf{0}$. We compute this minimum using a projected gradient method, which iteratively takes steps in the direction of the negative gradient and projects the result onto the feasible region to ensure repulsive forces.

At the optimal solution $\boldsymbol{\gamma}^*$, the Karush–Kuhn–Tucker (KKT) conditions are satisfied. These constitute the first-order optimality conditions for constrained problems, extending the stationarity condition $\nabla E = 0$ to include inequality constraints via Lagrange multipliers. Under convexity, they are both necessary and sufficient for global optimality [25]. In particular, primal feasibility, dual feasibility, and complementary slackness jointly characterize the solution. Importantly, the KKT conditions admit a direct correspondence with the imposed physical constraints.

1. **Primal feasibility** ($\boldsymbol{\gamma}^* \geq 0$): We enforce this explicitly through a projection (`max` operator) during minimization, which prevents forces from becoming attractive. Physically, this ensures that contact forces only push cells apart according to the **Repulsive Forces** constraint.

2. **Dual feasibility** ($\nabla E(\boldsymbol{\gamma}^*) \geq 0$): At the solution $\boldsymbol{\gamma}^*$, the directional derivative of the energy must be non-negative in every feasible direction. A negative partial derivative would indicate that the energy could be reduced by increasing γ_α, contradicting the optimality of $\boldsymbol{\gamma}^*$. Given that $\nabla E(\boldsymbol{\gamma}^*) \approx \mathbf{\Phi}^{k+1}(\boldsymbol{\gamma}^*)$ (from Equation 14), this condition ensures $\mathbf{\Phi}^{k+1} \geq 0$, enforcing the **No-Overlap** constraint.

3. **Complementary slackness** ($\boldsymbol{\gamma}^{*\top} \mathbf{\Phi}^{k+1} = 0$): Since both terms are non-negative, each contact must satisfy either $\gamma_\alpha = 0$ (no force without contact) or $\Phi_\alpha^{k+1} = 0$ (force can be non-zero with contact). Physically, forces act only where cells are actually touching, satisfying the **Complementarity Condition**.

Thus, the unique global minimizer $\boldsymbol{\gamma}^*$ resolves all contacts according to the physical laws of rigid-body interactions. This equivalence between the KKT conditions of a convex optimization problem and a complementarity problem is widely used in constraint-based physics simulations [15, 23, 24, 28, 29, 32, 36, 39, 41].

4.5.5. Numerical Solution

We solve the constrained optimization problem using the Barzilai-Borwein projected gradient descent method (BBPGD) [10], following [36, 41]. BBPGD is an iterative first-order method for box-constrained energy minimization. Each iteration performs a gradient descent step on $E(\boldsymbol{\gamma})$ followed by a projection onto the feasible set $\boldsymbol{\gamma} \geq \mathbf{0}$ to enforce non-negativity of contact forces. The step-size is determined using the Barzilai-Borwein rule, which provides an inexpensive approximation to second-order curvature information without line searches.

The iteration is continued until convergence, yielding optimal contact forces $\boldsymbol{\gamma}^*$. These are then used to compute $\boldsymbol{\mathcal{F}}(\boldsymbol{\gamma}^*)$ and $\dot{\boldsymbol{\ell}}(\boldsymbol{\gamma}^*)$ for updating the colony state via Equation 6.

Following Weady et al. [36], we adopt a convergence criterion based on a residual that directly measures physical overlap. The solver terminates when this residual falls below a user-specified tolerance of $\varepsilon = d/500 = 0.5/500 = 10^{-3}$, ensuring that the maximum overlap between any pair of cells remains below 0.1% of the cell diameter $d = 0.5$.

4.5.6. Limitations of the Linearization

A fundamental limitation of the linearization is that orthogonal motions, specifically rotations about contact points and translations parallel to the contact plane, are unconstrained within the linear approximation [36]. Consequently, even after the BBPGD algorithm converges to the specified tolerance ε, residual overlaps may still be present that violate the non-overlap constraint.

To address this limitation, Weady et al. [36] developed the recursively generated linear complementarity problem (ReLCP) method. After each BBPGD convergence, ReLCP identifies any remaining overlaps by examining all pairwise cell distances, generates new constraints for these violations, and re-solves the complementarity problem with the augmented constraint set. This process iterates until all cell overlaps fall below the tolerance ε. This hierarchical approach ensures that the non-overlap condition is strictly enforced, even in the presence of complex contact geometries and coupled motions (see Figure 19b). The iterative refinement procedure typically converges within a small number of iterations, making it computationally tractable for most simulations [36].

Nevertheless, this approach still requires a sufficiently small step-size, $\Delta t \sim 2 \cdot 10^{-3}$ (see Figure 18), to prevent excessive cell motion [39]. If cells move too far in a single timestep, new overlaps can form that the linearization does not capture, preventing constraint resolution. In practice, this can cause the BBPGD method to fail to converge or the ReLCP solver to be unable to resolve all overlaps within a reasonable number of iterations, leading to simulation failure.

Despite this computational restriction, the hard model's step-size requirement is less severe than the soft model's, though it still represents a significant computational bottleneck for large colonies with millions of cells.

5. Implementation

All simulations in this paper were performed with a custom C++ framework designed for large-scale simulations of proliferating cell collectives. The framework leverages distributed-memory parallel computing and standard high-performance computing (HPC) libraries to achieve scalability, enabling simulations of hundreds of thousands of cells across hundreds of CPU cores. The implementation combines concepts from Weady et al. [36] and similar frameworks [32, 41].

5.1. Distributed Computing Architecture

The core architecture is built on two foundational libraries:

- **PETSc** [2]: Used as a backend for distributed vectors and sparse matrices, supporting distributed computation across hundreds of CPU cores via MPI. PETSc partitions data across MPI processes and transparently handles inter-process communication for global operations (e.g., sparse matrix-vector products, iterative solvers, and global reductions).

- **MPI**: Underlies all inter-process communication, including ghost-cell exchanges (copies of boundary particles (i.e. cells) replicated to neighboring MPI ranks for cross-boundary collision detection) at domain boundaries to maintain consistent collision handling across partitions.

Built on these abstractions, the framework implements the full physics pipeline, including both the ReLCP solver for the hard collision model and the soft collision model.

5.2. Collision Handling Pipeline

Collisions are processed through a unified pipeline designed to decouple geometry from physics. The process consists of three phases: (1) broad-phase detection using a spatial grid (reducing neighbor searches to $O(N)$), (2) narrow-phase detection that computes exact contact points, normals, and lever arms, and (3) force and stress assembly. The resulting contact geometry is passed to either the hard or soft collision model, which computes the per-contact forces and torques that are accumulated into the global force vector and used to update the colony stress state. This abstraction allows seamless switching between collision models without altering the overall simulation flow.

Each collision detector call generates a single constraint per neighboring cell-pair within a threshold distance $d = 0.5$, equal to one cell diameter, ensuring both overlapping cells and cells in near-contact are captured for constraint generation. This over-detection is crucial for the stability of the hard model, where near-contact pairs provide global context that improves constraint resolution. Ideally, constraints would be constructed between all cell pairs regardless of distance, but for computational efficiency only nearby cells are considered [39, 40, 41]. For the soft model, this over-detection does not apply: only truly overlapping pairs ($\delta \geq 0$) generate constraints, so near-contact but non-overlapping pairs contribute neither forces nor constraint objects.

5.3. Adaptive Timestepping

To enhance stability and convergence, particularly for fast-moving cells that might overlap significantly within a timestep, we employ an adaptive timestepping scheme, allowing the step-size Δt to vary dynamically based on the current state of the colony. To our knowledge, this approach has not been implemented in comparable engines.

The adaptive timestep strategy (outlined in Algorithm 1) is based on the Courant-Friedrichs-Lewy (CFL) condition [8], which provides a stability criterion linking temporal and spatial resolution:

$$\mathrm{CFL} = \frac{u_m \, \Delta t}{\Delta x} \leq c_{\mathrm{CFL}}. \tag{18}$$

~~For~~ where, for our cell-based system, ~~we define a~~ c_{CFL} is the CFL safety factor, u_m is the characteristic velocity scale ~~u_m~~, defined as the median of ~~$u_i = \|v_i\| + \dot{\ell}_i$~~ $u_i = \|\|\|\mathbf{u}_i\|\|\|_2 + \dot{\ell}_i$, where $\|\|\| \cdot \|\|\|$ is the L2-norm, over all cells, accounting for both mechanical velocities and growth-induced elongation. The characteristic spatial scale Δx is the solver tolerance $\varepsilon = 10^{-3}$, which controls the maximum allowable displacement per step. Enforcing that cells move no more than a fraction of ε per timestep leads to the following expression for the adaptive step-size:

$$\Delta t = \frac{c_{\mathrm{CFL}} \cdot \varepsilon}{u_m} \tag{19}$$

~~where~~ with $c_{\mathrm{CFL}} = 0.5$ ~~is the CFL safety factor~~ (ensuring cells move at most half the tolerance per step), and where $\varepsilon = 10^{-3}$ is the solver displacement tolerance~~, and u_m is the median particle speed including growth as defined above~~. Choosing Δt this way ensures that most cells move less than half the solver tolerance in each timestep, preventing excessive overlaps that could destabilize the simulation.

Algorithm 1 Adaptive Timestep Control

Input: Particle set $\{p_i\}$ with velocities v_i and growth rates $\dot{\ell}_i$, current step-size Δt, solver tolerance ε, smoothing factor $\cancel{\alpha}\beta$.
Output: Updated step-size Δt.
Compute characteristic velocity $\cancel{u_i = \|v_i\| + \dot{\ell}_i}$ $u_i = \|\|\mathbf{u}_i\|\|_2 + \dot{\ell}_i$ for each particle.
Calculate u_m as the median of $\{u_i\}$.
Determine Δt^* using Equation 19.
Smooth the step-size change: $\cancel{\Delta t_{\text{new}} = (1-\alpha)\Delta t + \alpha \Delta t^* \text{ with } \alpha = 0.01}$ $\Delta t_{\text{new}} = (1-\beta)\Delta t + \beta \Delta t^*$ with $\beta = 0.01$.
Clamp Δt_{new} within 20% of current Δt to prevent abrupt changes.
Update simulation step-size: $\Delta t \leftarrow \Delta t_{\text{new}}$.

The smoothing factor $\cancel{\alpha = 0.01}$ $\beta = 0.01$ and the 20% clamping threshold are introduced to regularize step-size updates and prevent abrupt fluctuations in Δt. This damping mechanism suppresses high-frequency oscillations that may arise from transient changes in the velocity distribution, while preserving responsiveness to gradual shifts in colony dynamics.

We base u_m on the median rather than the maximum cell speed to prevent a few fast-moving cells from dictating the step-size for the entire colony. Since $\Delta t = c_{\text{CFL}}\, \varepsilon / u_m$ (Equation 19) is applied globally, a maximum-based criterion would be set by the single fastest cell (for instance, one transiently accelerated during a local rearrangement), forcing every cell in the colony onto a smaller step-size and slowing the simulation. The median instead produces a step-size suitable for the bulk motion that governs colony expansion.

5.4. Simulation Output and Availability

The full framework is openly available at `https://github.com/manuellerchner/MicrobeGrowthSim-IDP` and includes:

- **Simulation**: Both hard and soft collision models in a shared C++ codebase with MPI parallelization and adaptive timestepping. Simulation state is exported in Visualization Toolkit (VTK) format at user-defined intervals, recording positions, orientations, lengths, stresses, growth rates, and contact forces for visualization in ParaView [1].
- **Analysis**: Jupyter notebooks for radial distribution profiles (packing fraction, stress, growth rate, cell length), microdomain cluster analysis, BBPGD convergence diagnostics, and energy landscape visualization.
- **Benchmarking**: Python scripts and notebooks for strong-scaling studies, CFL factor sweeps, and colony-size runtime profiling, reproducing all performance results in this work.
- **Utilities**: Modules for parsing VTK output, loading and aligning multi-run datasets, generating simulation configurations, and rendering colony snapshots via ParaView state files.

6. Pattern Formation Analysis

With both models sharing identical growth, stress computation, and timestepping infrastructure, we now compare their emergent behavior across a range of observables spanning pattern formation and colony-scale mechanics. In particular, we examine concentric ring patterns, cell density and local packing fraction, colony growth dynamics and total cell number, microdomain formation, and the coupled radial stress and growth rate profiles. This systematic comparison is designed to isolate the effect of the collision model on macroscopic behavior, and to determine whether differences manifest primarily at the level of local structure, global growth dynamics, or stress-mediated feedback. We further identify parameter regimes in which both formulations produce consistent results, as well as regimes where their predictions diverge.

6.1. Concentric Ring Patterns

We validate both implementations by reproducing concentric ring patterns that arise under stress-sensitive growth, as reported in [35]. As shown in Figure 4, both models successfully produce qualitatively similar ring structures, demonstrating that the core mechanics of growth and mechanical feedback are effectively captured by both collision handling approaches.

The effect of λ on pattern formation is consistent across both models. At low stress-sensitivity (e.g., $\lambda = 10^{-4}$, corresponding to weak growth inhibition), stress gradients are too weak to generate significant spatial patterns, and thus no rings form. At intermediate sensitivity (e.g., $\lambda = 10^{-3}$), distinct concentric rings emerge early in the simulation and persist as the colony expands, with ring spacing determined by the balance between growth and stress. At high sensitivity ($\lambda = 10^{-2}$, corresponding to strong growth inhibition), an initially formed ring quickly dissipates as cells reorient to random orientations and growth nearly ceases due to strong mechanical inhibition throughout most of the colony.

Despite these qualitatively similar outcomes, visual differences emerge between the models. The soft model produces visibly denser patterns with a fuzzy appearance due to allowable cell overlap, while the hard model yields sharper, more clearly defined rings. Nevertheless, macroscopic features, such as the number of rings, spacing, and colony expansion rate, remain comparable between the two models, as shall be shown quantitatively.

6.2. Cell Density and Local Packing Fraction

The most striking difference between the models is the cell packing density (see Figure 6). The soft model exhibits significantly denser packing in the colony center, decreasing radially outward (see Figure 7a). For very low stress-sensitivity ($\lambda = 10^{-4}$), local packing fractions (defined as the ratio of cell area to available area) reach values of 5, indicating severe cell overlap. Values exceeding 1.0 are

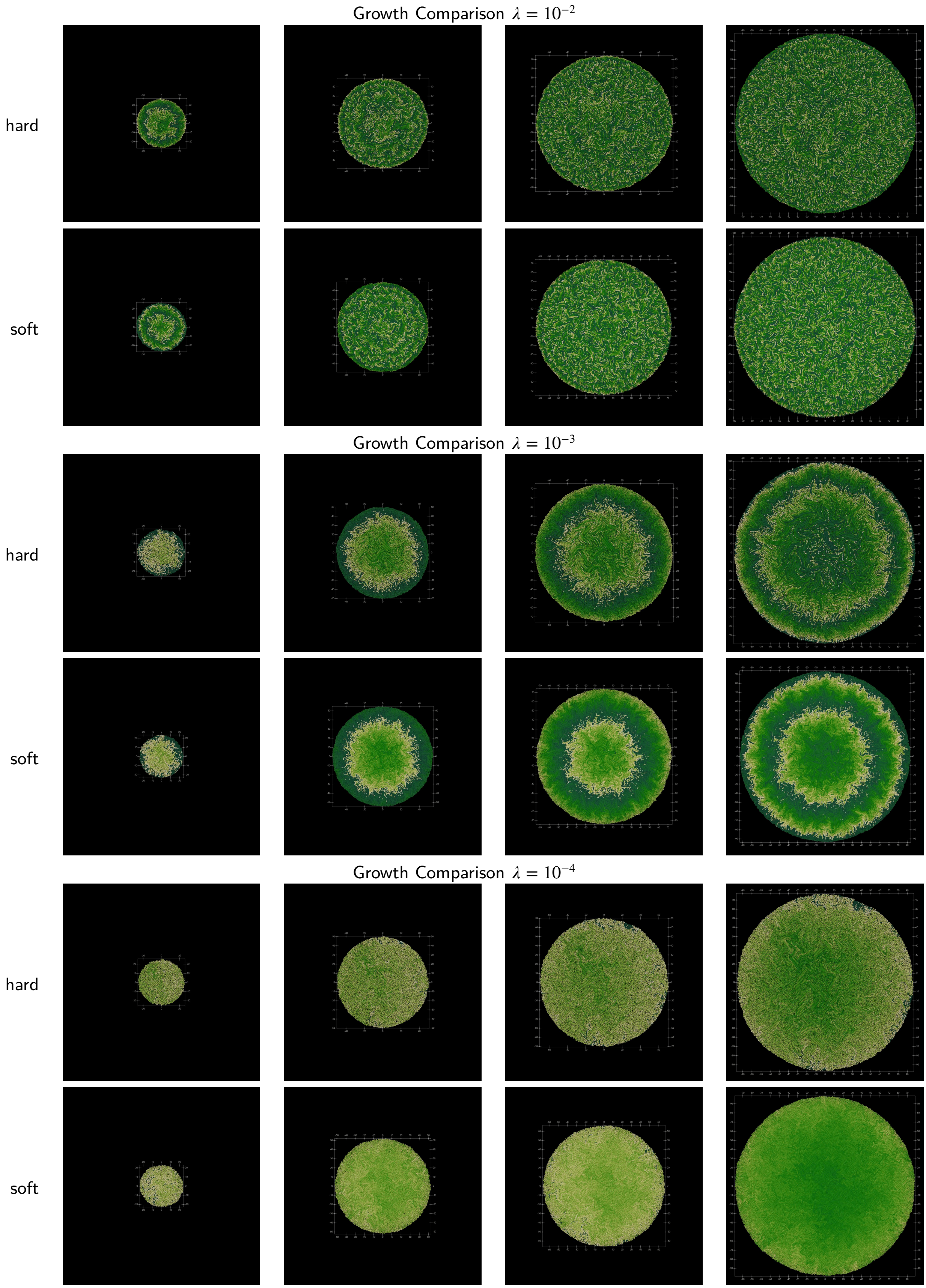


Figure 4: Comparison of pattern formation between hard and soft collision models at similar colony sizes. Each row shows the colony evolutions up to a maximum colony radius of 100. The color indicates the length of the cells (short cells are darker, longer cells are lighter). Concentric ring patterns are visible for both models at $\lambda = 10^{-3}$ confirming the results from [35].

impossible for truly rigid spherocylinders and reflect the soft model's inability to prevent interpenetration. In contrast, the hard model maintains approximately constant packing density of 0.9 throughout the colony, consistent with densely packed non-overlapping rods.

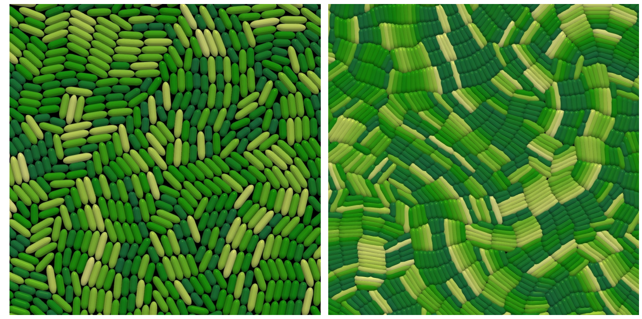
The unphysical packing fraction ~~arises because the soft model relies on~~ is primarily a consequence of the finite stiffness of the repulsive potential. To show this, we perform a controlled study of the impact of increasing k_{cc} on growing colonies to radius $R \approxeq 20$, measuring the impact this has upon the radial packing fraction, defined as $\phi(r) = \frac{A_{\text{cells}}(r)}{A_{\text{total}}(r)}$, where r is the radius, $A_{\text{cells}}(r)$ is there combined area of all cells within radius r and $A_{\text{total}}(r)$ the area of the bounding circle of radius r. In particular, we look at the center packing fraction $\phi_{\text{center}} = \phi(6)$.

We see that with an increased contact stiffness k_{cc}, it is possible to reduce the central packing fraction below the physically-realistic $\phi = 1$ limit and converge towards the $\phi \approx 0.9$ packing fraction of the hard model (Figure 5a). This cannot be obtained for free, since the critical CFL factor at which the simulation is stable is inversely proportional to the stiffness k_{cc} (Figure 5b). Increasing k_{cc} to produce similar packing fractions to the hard model will therefore come with computational costs increasing by over an order of magnitude. This shows that the unphysical packing fraction of the soft model is primarily a result of the finite stiffness of the repulsive model, although a possible secondary contributing factor could be that the local pairwise forces ~~that~~ cannot fully propagate the accumulated stress to the colony edge~~. Combined with a numerical step-size too large to resolve all collisions, this leads to excessive cell overlap and overcrowding in the interior.~~, leaving the interior overcrowded.

For the remainder of this work, we will continue with a contact stiffness of 2×10^4, as defined by some sources in the literature [6, 43], although these results will impact our conclusion of the comparison.

Figure 7b quantifies this effect: the soft model exhibits substantially elevated cell overlap at all distances from the colony center, approaching full cell overlap near the colony boundary. In contrast, the hard model maintains maximum overlap near the solver tolerance $\epsilon = 10^{-3}$ at every radial position, confirming that the strict non-overlap constraints are enforced globally.

~~A contributing factor to this excessive overlap is the adaptive timestep method~~ The adaptive timestep (see Algorithm 1) ~~, which~~ bases Δt ~~solely on cell velocities but not on current overlap. A more sophisticated scheme could consider maximum overlap when adjusting Δt, however~~ on the median cell velocity. A criterion based on the maximum cell velocity, a high percentile of the velocity, or an overlap-aware criterion could suppress such cell interpenetration by resulting in smaller step-sizes~~would come with reduced performance~~. However, this would come at increased computational cost for the soft model.



(a) Hard model (b) Soft model

Figure 6: Close-up comparison of cell packing in the colony center for both collision models. The hard model (a) maintains near-optimal packing, while the soft model (b) exhibits significant overlap and overcrowding.

6.3. Colony Growth Dynamics and Number of Cells

A direct consequence of the overcrowded interior in the soft model is its effect on the total number of cells. The soft model requires substantially more cells to reach the same colony radius (see Figure 8a). In particular, when stress-sensitivity is low ($\lambda = 10^{-4}$), it needs almost three times as many cells as the hard model to achieve the same radius. This disparity shows that the densely packed interior contributes little to outward colony pressure: interior cells remain largely immobile yet continue to grow and divide, creating a large excess of cells that do not contribute to expansion.

Figure 8b compares simulation time and colony radius for both models. Despite the difference in cell numbers, their overall growth dynamics are similar, with the soft model growing slightly more slowly due to cell overlap and with smaller λ resulting in faster growth for both models. At high stress-sensitivity ($\lambda = 10^{-2}$), both models transition from early exponential growth to stress-limited linear growth as mechanical stress accumulates in the colony interior.

6.4. Radial Stress Distribution and Growth Rate

Another key difference between the two modeling approaches lies in the radial stress distribution that develops within the colony (see Figure 9). Both the hard and soft models produce qualitatively similar radially averaged stress profiles, consistent with the behavior originally reported in [35]. In both cases, the stresses are highest at the colony center and gradually decrease toward the outer edge, reflecting the buildup of mechanical compression in the densely packed interior. These profiles approximately follow the ~~derived analytical expression~~ $\cancel{\tilde{\sigma}(r) \approx \frac{2}{\lambda}\ln\left(\frac{1}{8c} - c\lambda r^2\right)}$ analytical expression for the stress for the continuum model derived in Weady et al. $\tilde{\sigma}(r) \approx \frac{2}{\lambda}\ln\left(\frac{1}{8c(R,\lambda)} - c(R,\lambda)\lambda r^2\right)$, where $c(R,\lambda) = \frac{(1+\lambda R^2/2)^{1/2}-1}{2\lambda R^2}$, for colony radius R [35], predicting a logarithmic decay of stress with distance r from the colony center.

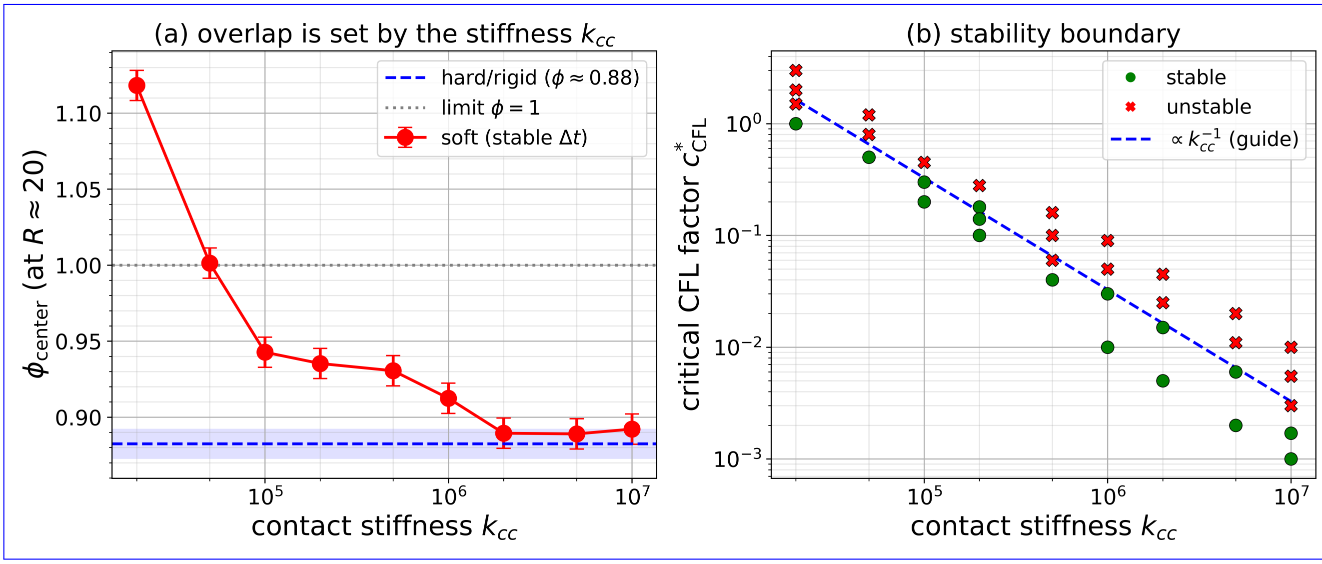


Figure 5: The soft model's excess central packing results from the contact stiffness, but reducing this to produce similar packing fractions as the hard model requires a lower c^*_{CFL} and hence larger step-size. **(a)** The overlap is set by the stiffness: increasing k_{cc}, each run at a step-size small enough for stability, drives the center packing fraction $\phi_{\text{center}} = \phi(6) = \frac{A_{\text{cells}}(6)}{A_{\text{total}}(6)}$ (compared at $R \approx 20, \lambda = 10^{-4}$) toward the rigid packing of the hard model (dashed). **(b)** The stiffness cannot be increased for free. The critical CFL factor c^*_{CFL} separating stable (green) from unstable (red) runs scales at approximately k_{cc}^{-1}. A run is classified as unstable if any cell's single-step displacement exceeds half its length. A soft model (large k_{cc}) that closely reproduces the hard model therefore requires a prohibitively small step-size and thus a higher cost.

Although the overall shape of the profiles is comparable, the magnitude of the predicted stresses differs substantially between the two models. Quantitatively, the hard model produces stress values that are up to two times higher than the analytical prediction, whereas the soft model yields stresses that are only about half the analytical value. This variation in stress magnitude is significant and likely plays a central role in explaining the discrepancies in growth dynamics and microdomain organization discussed in subsection 6.3 and subsection 6.5.

The consistently elevated stresses observed in the hard model are also noted by Weady et al. [35] and are likely a consequence of the discrete cell-based representation not fully satisfying the continuum assumptions used in the analytical derivation.

In contrast, the soft model produces lower stresses because it lacks a global force propagation mechanism: stress is transmitted only through local pairwise repulsive interactions, which are insufficient to communicate mechanical load effectively across the entire colony. Consequently, stresses in the colony interior remain low, and central cells experience weaker mechanical inhibition. These cells therefore continue to grow and divide, eventually leading to the overcrowding and distorted microdomain patterns.

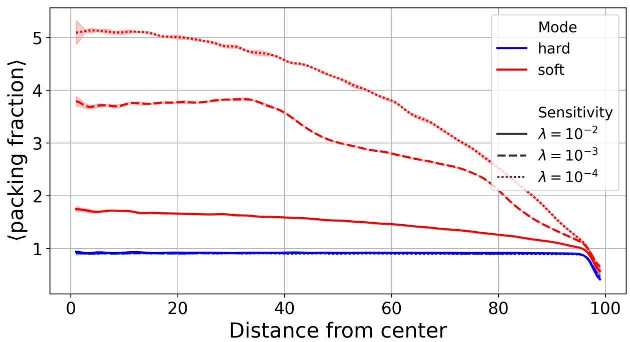
As shown in Figure 10, the differences in stress transmission directly manifest in the radial growth rate profiles. Cells in the soft model exhibit substantially higher growth rates than those in the hard model across most of the colony interior, particularly at low stress-sensitivity values. At higher stress-sensitivity ($\lambda = 10^{-2}$), growth becomes strongly suppressed in the colony interior for both models, and only cells located near the periphery remain actively dividing. The resulting shift from interior-driven to edge-driven proliferation is also clearly visible in Figure 8b and mirrors a well-documented behavior in real microbial systems, such as in bacterial and yeast colonies, including *E. coli* and *Saccharomyces cerevisiae*, where both mechanical feedback and nutrient depletion drive this transition [18, 20, 34]; our model isolates the mechanical contribution alone.

6.5. Microdomain Formation

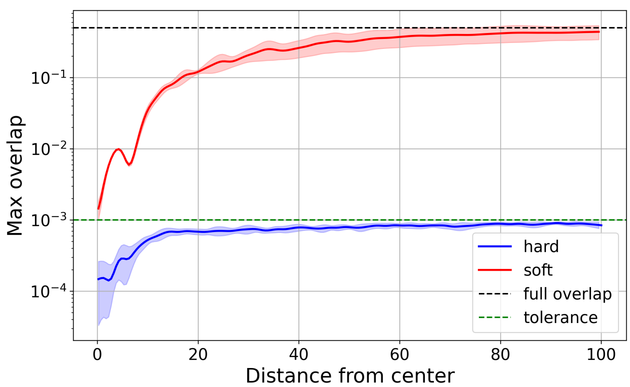
Similar to You et al. [42, 43], we observe the formation of microdomains, defined as localized regions where cells share similar orientations. These domains emerge from mechanical interactions and growth dynamics and result in mosaic-like patterns of cell alignment within the colony.

As shown in Figure 11, both hard and soft collision models reproduce this phenomenon for small colonies ($R \approx 50$), producing patterns that closely resemble experimental observations of real *E. coli* colonies [43]. This agreement suggests that at small scales, where the soft model's packing artifacts remain limited, the key features of microdomain formation can be captured through both collision handling approaches.

Although the hard and soft collision models presented here are not directly comparable to You et al. [43], because their study assumes constant growth rates while our models use stress-dependent growth, their findings still provide useful context. You et al. show that reduced growth rates lead to larger microdomains. In our models, stress sensitivity λ

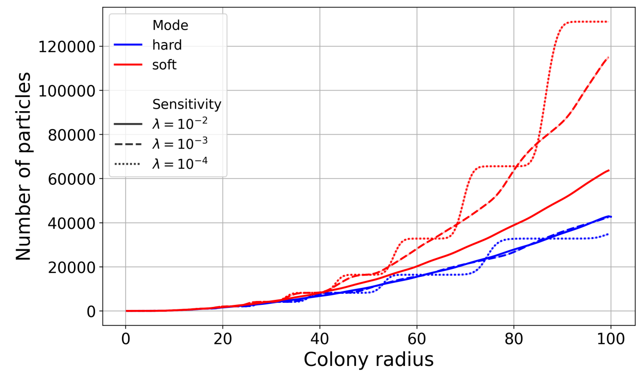


(a) Radial packing fraction profiles ($\phi = \frac{A_{\text{cells}}}{A_{\text{total}}}$) for both models. Here, A_{cells} denotes the total area occupied by cells within a spatial bin of width 2 at distance r from the colony center, and A_{total} is the corresponding annular bin area. The hard model maintains nearly optimal packing, while the soft model exhibits elevated fractions in the colony center, particularly at low stress-sensitivity ($\lambda = 10^{-4}$).

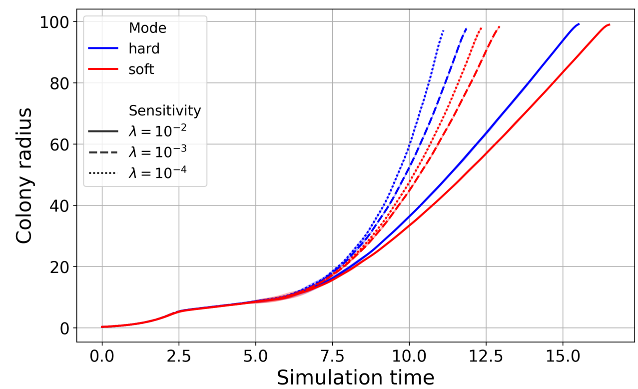


(b) Maximum cell overlap as a function of distance from the colony center. The hard model remains uniformly near solver tolerance $\varepsilon = 10^{-3}$, whereas the soft model exhibits substantially elevated overlap at all radial positions, with values near the colony boundary approaching full cell overlap.

Figure 7: (a) Radial packing fraction profiles. (b) Maximum cell overlap vs. distance from colony center. Both graphs study a colony of radius $R \approx 100$.



(a) Number of cells required to reach a given colony radius. The soft model requires substantially more cells, particularly at low stress-sensitivity ($\lambda = 10^{-4}$).



(b) Simulation time vs. colony radius. Both models exhibit similar growth dynamics, though the soft model is slightly slower due to allowable cell overlap.

Figure 8: Colony growth dynamics and cell scaling comparison between hard and soft models. (a) Number of cells vs. colony radius. (b) Colony radius vs. simulation time.

produces a similar effect by inhibiting growth in the densely packed colony center, resulting in larger microdomains. This trend is evident in simulations of the soft model (see Figure 12), where higher λ values correspond to slower growth and larger microdomains, consistent with You et al.'s observations. The hard model, in contrast, shows no such trend. The consistent microdomain size across λ values in the hard model may be related to the global constraint resolution mechanism, but a detailed investigation of the underlying mechanisms is beyond the scope of this comparison.

Both models produce similar orientation patterns in small colonies at a colony radius around $R \approx 50$ (see Figure 11). However, pronounced differences emerge as colonies grow larger or λ decreases. As shown in Figure 12, the hard model maintains distinct patches of similarly-sized aligned cells across all λ values. In contrast, the soft model produces elongated, unrealistic bundles of densely packed aligned cells that do not resemble experimentally observed microdomains [43].

This artifact stems directly from the soft model's higher interior growth rates, discussed in the previous section. Without sufficient outward stress propagation, the interior becomes unphysically dense, forcing cells into long, aligned bundles.

The effect of λ on microdomain structure is particularly apparent in larger colonies. As shown in Figure 13, the hard model produces microdomains of fairly uniform size across all stress-sensitivity values, while the soft model produces larger microdomains with higher variability, as λ decreases.

To complement the domain-size analysis with a measure of the *local* orientational order, we compute the nematic orientation correlation function $C(r) = \langle \cos(2\Delta\theta) \rangle$, where $\Delta\theta$ is the angle of between two cells [31], over all cell pairs with centers separated by a distance r, using only cells in the colony core (within 80% of the colony radius away from the colony center) for moderately sized colonies, $R \approx 48$ (Figure 14). This metric describes the relationship in orientation of the rods at different distances, with $C(r) = 1$

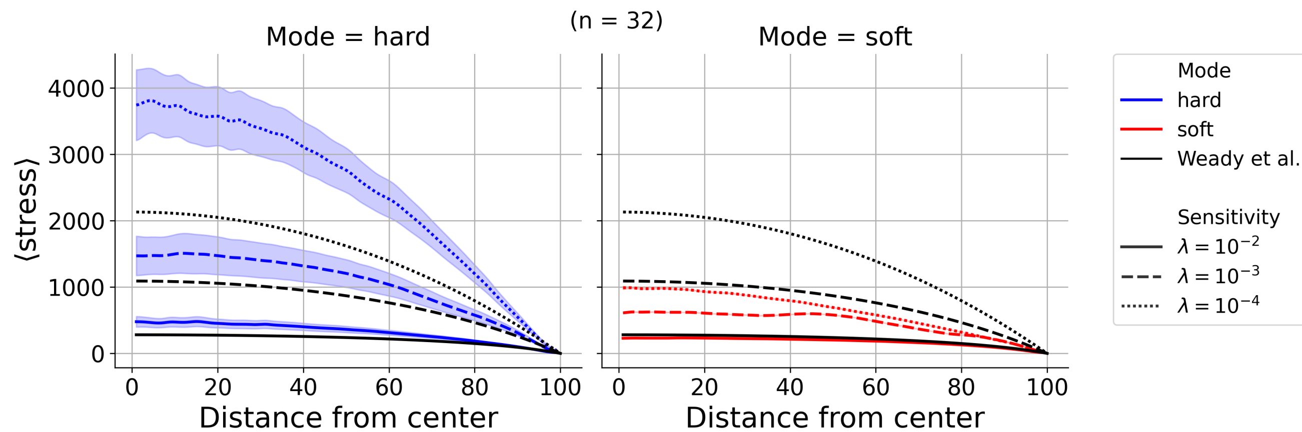


Figure 9: Radial stress profiles for both models across different λ values. Both models exhibit similar qualitative trends, with stresses peaking at the colony center and decreasing toward the edge. However, the hard model consistently produces higher stress magnitudes compared to the soft model. The black curves represent the analytical prediction from [36], shown for each λ value with the corresponding line style.

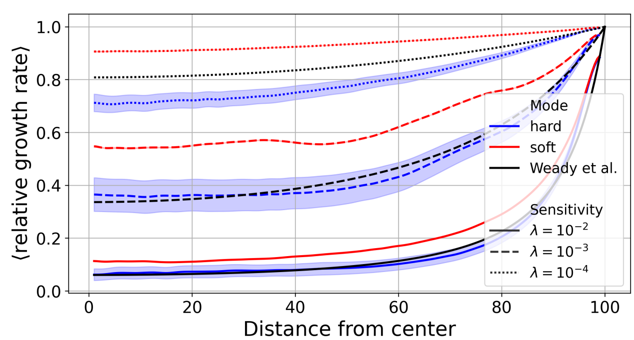
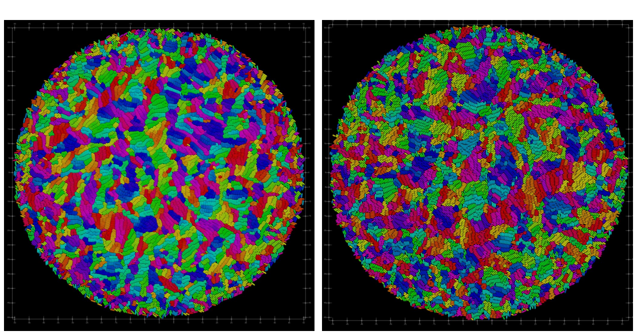


Figure 10: Radial relative growth rate profiles ($e^{-\lambda\sigma}$) for both models. Cells in the soft model grow significantly faster than those in the hard model across most of the colony, particularly at low stress-sensitivity values. At high stress-sensitivity ($\lambda = 10^{-2}$), growth is strongly suppressed in the interior of both models, with only cells at the colony edge actively dividing. The black lines indicate the theoretical growth rates derived from the continuum model in [35].



(a) Soft model with $\lambda = 10^{-2}$ at colony radius $R \approx 50$.

(b) Hard model with $\lambda = 10^{-2}$ at colony radius $R \approx 50$.

Figure 11: Cell orientation patterns for both models. The color indicates cell orientation angle, with similar colors representing similar angles. Both models produce comparable microdomain structures in small colonies.

indicating that all cells at that distance lie parallel to each other, $C(r) = -1$ indicating they lie perpendicular, and $C(r) = 0$ indicating no relationship in orientation. We see for both models and all stress sensitivity parameters, λ, a decay from $C(0) = 1$ to $C(r) \to 0$ as $r \to \infty$, showing a trend from typically parallel alignment at close distances to a lack of alignment at larger distances.

To get the nematic orientation correlation length ξ_θ, we fit exponential functions $g(r) = \exp\left(-\frac{r}{\xi_\theta}\right)$ on each orientation correlation function (Table 1 [9]). In all cases $C(r)$ decays to $1/e$ within one to two cell lengths, defining an orientational correlation length ξ_θ that grows with stress sensitivity, consistent with stronger growth inhibition suppressing the rearrangements that scramble local alignment. Because ξ_θ stays on the order of a single cell length, any local orientation correlation rarely survives more than a few cell widths. Critically, ξ_θ is largely comparable between the hard and soft models at every λ, so the differences between the models reside more in the larger-scale domain organization (Figure 13) rather than in local alignment.

Table 1
The nematic orientation correlation lengths ξ_θ, obtained by fitting exponential decay functions $g(r) = e^{\frac{r}{\xi_\theta}}$ on the orientation correlation functions in Figure 14. The orientation correlation lengths are all within one or two cell lengths, with larger λ resulting in greater lengths. There are no significant differences between the two models, with slightly greater lengths for the soft model at $\lambda = 10^{-3}$ and $\lambda = 10^{-2}$ and greater length for the hard model at $\lambda = 10^{-4}$.

	$\lambda = 10^{-4}$	$\lambda = 10^{-3}$	$\lambda = 10^{-2}$
Hard	1.31	1.40	1.89
Soft	1.19	1.45	1.94

As a consequence of bundle formation and overcrowding, the soft model is not suitable for studying microdomain

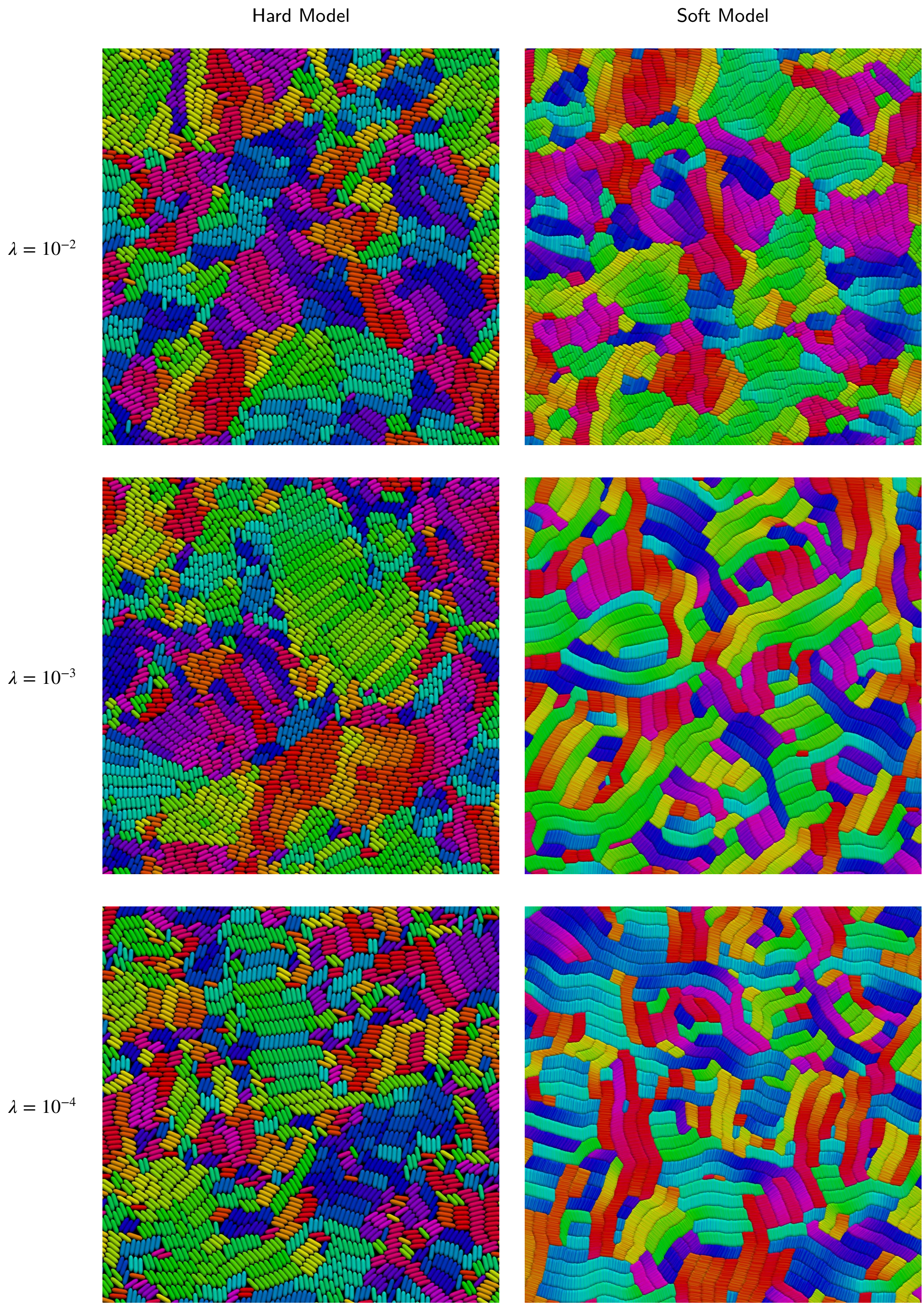


Figure 12: Comparison of orientation patterns at different stress sensitivities in the center of the colony. The color indicates the cell's orientation, with similar colors representing similar angles. The hard model is able to maintain visible patches of aligned cells for all stress sensitivities, whereas the soft model tends to produce long bundles of densely packed cells for low stress sensitivities. This effect is caused by a lack of separation forces, causing cells in the center of the colony to pile up.

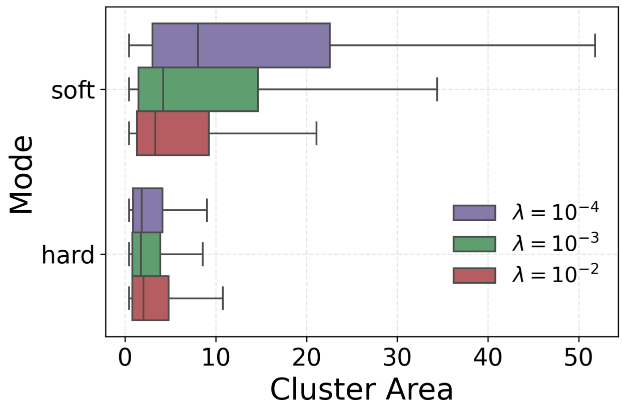


Figure 13: Microdomain area distributions for large colonies ($R \approx 100$) across λ. The hard model yields consistent domain sizes, while the soft model exhibits greater variability with larger clusters, particularly at $\lambda = 10^{-4}$. Clusters are identified via a graph-based connected-component method [43], grouping neighboring cells with similar orientations.

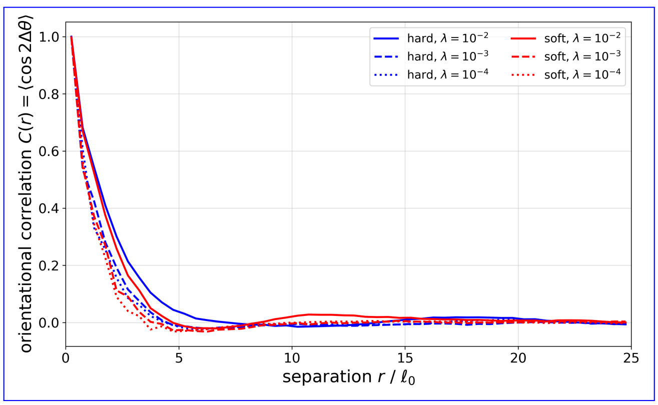


Figure 14: Nematic orientational correlation function $C(r) = \langle\cos(2\Delta\theta)\rangle$ for both models at three stress sensitivities, measured in the core of colonies of radius $R \approx 48$, with bin widths of 0.5 and using only cells within 80% of the colony radius away from the center. Color denotes the collision model (blue: hard, red: soft) and line style denotes the stress sensitivity λ. The correlation exponentially decays at a rate of one to two cell lengths in all cases.

formation in large colonies or in regimes of low stress sensitivity, where these effects distort the local dynamics and lead to unrealistic structural organization. In such cases, excessive overlap and artificial clustering interfere with the natural development of orientational order, preventing a faithful representation of the physical mechanisms driving microdomain formation.

For smaller colonies ($R \lesssim 50$) and at high stress-sensitivity values, the soft model's packing artifacts remain somewhat contained and the overall orientation patterns show qualitative agreement with the hard model. Within these limited conditions, it may serve as a computationally efficient tool for preliminary exploration. For applications that require accurate characterization of microdomain properties or the exploration of weak stress-sensitivity regimes, the hard model remains the preferred choice.

7. Computational Performance

We now focus on the computational performance of both models. To leverage multiple CPUs, the simulation is parallelized using the Message Passing Interface (MPI), which allows the workload to be distributed efficiently across many cores. All benchmarks are carried out on the `CoolMUC-4` HPC cluster[2].

7.1. Domain Decomposition

For parallel execution, the simulation domain is divided into evenly spaced angular sectors of the circular colony, as illustrated in Figure 15. We introduce this angular sector decomposition specifically suited to the circular geometry of bacterial colonies. The center of the colony is exclusively assigned to rank 1 to avoid a singularity where all ranks would meet at the center. This decomposition ensures better load balancing for circular colonies than regular rectangular slices, which would produce significant load imbalance due to the colony's radial symmetry.

Communication between ranks occurs through ghost particle exchange at sector boundaries. At each timestep, particles near the boundaries are sent to neighboring ranks as ghost particles (copies of boundary particles replicated to neighboring MPI ranks for cross-boundary collision detection), allowing for correct collision detection across domain borders.

7.2. Strong Scaling

To assess parallel performance, we analyze runtime and speedup as functions of CPU core count. In the strong-scaling tests, colonies grow from a single cell to a radius of 100.

We find that the hard model consistently outperforms the soft model across all core counts, by a factor ranging from 1.20× at 1 core to 9.36× at 112 cores (see Figure 16a). This performance advantage likely stems from the hard model's significantly larger stable step-sizes, which allow more computational work per communication cycle. As shown in Figure 16b, the hard model achieves reasonable speedup up to approximately 64 cores, while the soft model scales well only up to about 32 cores.

The decrease in performance at higher ranks arises from the domain decomposition. At very high core counts, each rank handles only a narrow angular slice of the colony (see Figure 15), so communication increasingly dominates runtime. With less local work per rank, ghost particle exchanges and global PETSc matrix operations (in the hard model) become more frequent and costly, especially when

[2] `CoolMUC-4` is part of the Linux Cluster at the Leibniz Supercomputing Centre (LRZ). It consists of 106 compute nodes equipped with *Intel Xeon Platinum 8480+ (Sapphire Rapids)* CPUs, each providing 112 cores. For details, see `https://doku.lrz.de/coolmuc-4-1082337877.html`.

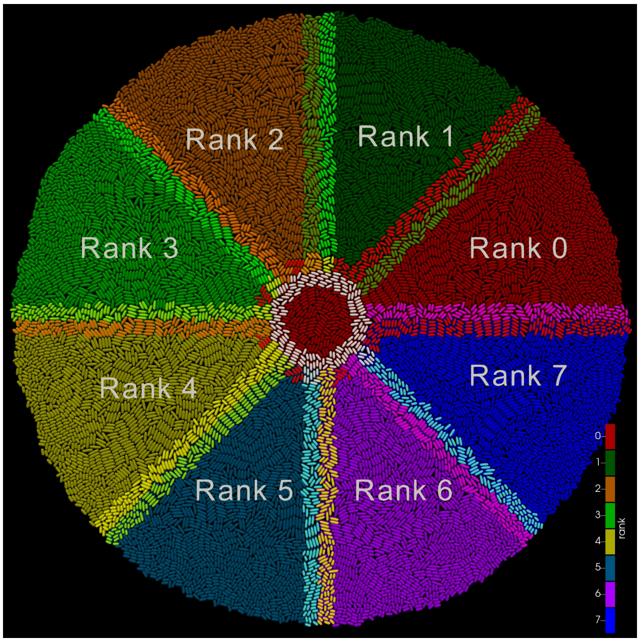
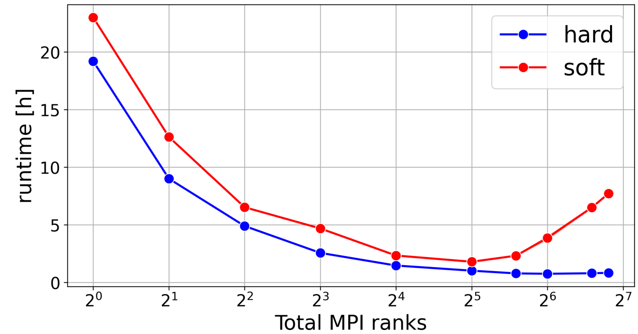
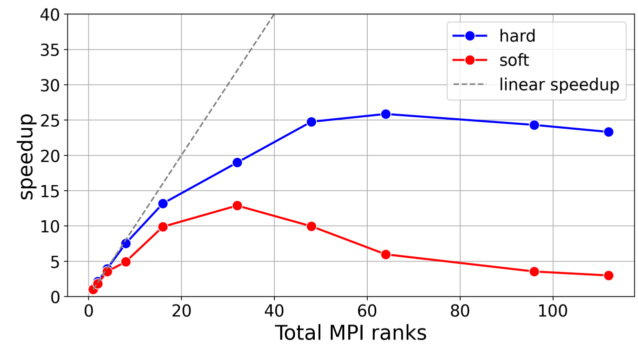


Figure 15: Domain decomposition for 8 MPI ranks showing evenly spaced angular sectors of a colony with radius 50. Each rank owns one angular sector. Ghost particles at rank boundaries are shown in brighter colors.



(a) Runtime to reach colony radius of 100 versus MPI ranks. The hard model is consistently faster, with the performance gap widening at high core counts.



(b) Speedup versus MPI ranks. The hard and soft models scale reasonably up to 64 and 32 ranks, respectively.

Figure 16: Strong scaling comparisons between hard and soft collision models. (a) Runtime vs. MPI ranks. (b) Speedup vs. MPI ranks.

small step-sizes Δt are used. This imbalance reduces scaling efficiency and leads to longer runtimes at high core counts.

7.3. Adaptive Timestep Analysis

A key factor in the soft model's poor parallel performance is its critically small step-size, determined by Equation 19. As the colony grows, Δt decreases sharply (see Figure 17), so each rank advances particles only minimally before triggering costly MPI communication (ghost exchanges and neighbor updates). These operations then dominate runtime, leading to poor scaling.

In contrast, the hard model maintains a mostly constant step-size that is significantly larger than that of the soft model. This allows each rank to advance particles substantially between communication steps, effectively amortizing the cost of MPI exchanges over more computational work. Consequently, the hard model sustains a favorable computation-to-communication ratio, explaining its better scaling at moderate to high core counts.

As shown in Figure 17, the hard model stabilizes around $\Delta t_{\text{hard}} \approx 3 \cdot 10^{-4}$, while the soft model steadily decreases to $\Delta t_{\text{soft}} \approx 10^{-5}$. The ratio $\Delta t_{\text{hard}} / \Delta t_{\text{soft}} \approx 35$ indicates that the hard model can take roughly 35 times larger steps than the soft model in dense colonies, requiring far fewer steps to simulate the same physical time. Notably, for the hard model with $\lambda = 10^{-2}$, the critical step-size even increases over time as interior cell growth slows (see Figure 10) and the overall median velocity u_m decreases.

Using adaptive timestepping provides both stability and efficiency by adjusting the step-size to the colony's current dynamics. A fixed step-size cannot achieve this balance: steps that are too small waste computational effort when interactions are sparse, while steps that are too large can lead to numerical instability in dense regions. This adaptivity enables long-term simulations to proceed efficiently without compromising accuracy or stability.

7.4. Impact of Adaptive Timestepping on BBPGD Performance

The efficiency of the BBPGD solver for the hard model also depends critically on the step-size. The step-size Δt determines both the number of BBPGD iterations per timestep as well as the total number of timesteps required for the simulation. This relationship produces a characteristic U-shaped runtime curve as a function of the CFL number and corresponding step-size, as shown in Figure 18.

The total runtime to reach a colony radius of 100 on 112 CPU cores exhibits a pronounced minimum at CFL ≈ 2, corresponding to cells moving roughly twice the solver tolerance ε per timestep. At small CFL values (CFL $\ll 1$), each timestep requires relatively few solver iterations due to accurate linearization (Equation 14) and minimal overlaps, but many timesteps are required, making the simulation inefficient. At large CFL values (CFL $\gg 3$), linearization assumptions break down as cells move too far per step,

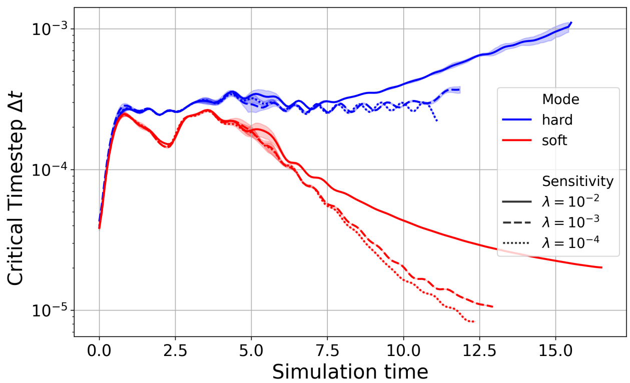


Figure 17: Adaptive step-size Δt over simulation time for both models. During early growth, Δt is similar and set by cell growth velocities. As the colony densifies, the soft model becomes collision-dominated, with large overlaps increasing velocities and reducing Δt to $\approx 10^{-5}$. The hard model avoids this, maintaining stable or increasing Δt due to constraint-based separation and reduced interior growth.

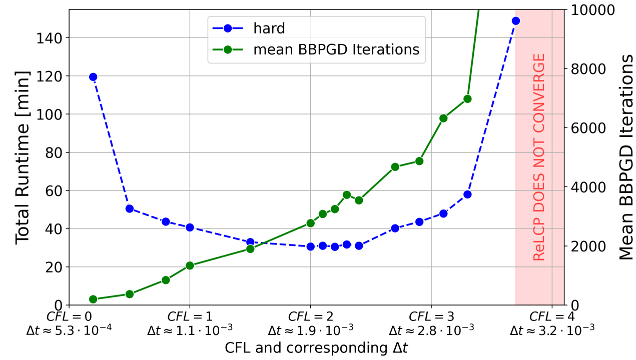


Figure 18: Total runtime to reach colony radius 100 versus CFL factor on 112 cores (each point is an independent run). A U-shaped trend emerges: small CFL values increase timestep count, while large values raise BBPGD iterations per step. The annotated Δt denotes the mean step-size. The red region indicates CFL values where the solver fails to converge.

increasing solver iterations, and in extreme cases (CFL > 3.7), the solver may fail to converge, as shown in Figure 18.

The average number of solver iterations per timestep increases with the CFL factor at low values, following iterations $\approx$ 1800 $\cdot$ CFL, reaching about 3600 near the optimum before rising sharply.

By selecting an appropriate CFL factor, adaptive timestepping maximizes physical progress per unit of computational effort, thereby minimizing total runtime. The step-size, determined by the CFL factor, represents a balance between relying on more constraint solver iterations per step and performing a larger number of total simulation steps. Optimizing this balance allows the hard model to achieve higher performance and improved scaling.

All benchmarks in the previous section used a conservative CFL factor of 0.5, indicating that runtimes could be improved even more by using a larger CFL value closer to the optimal point.

7.5. Insights into BBPGD Iterations

This section examines the performance of the BBPGD algorithm used in the hard collision model to solve the constrained optimization problem. Since the BBPGD algorithm is the most computationally demanding part of the simulation, understanding its efficiency and convergence is important. In colonies with radius $R \approx 100$, where numerous contact constraints must be satisfied, BBPGD and its associated matrix-vector operations account for roughly 85% of the total simulation time and therefore largely determine overall performance.

A key metric is the number of BBPGD iterations required per simulation step and how this quantity scales with system size. Figure 19a shows that iteration count remains mostly on the order of $\mathcal{O}(1000)$, with occasional peaks reaching $\mathcal{O}(5000)$ during critical moments such as large-scale rearrangements or near jamming transitions. This count includes additional BBPGD calls from the ReLCP procedure, which requires up to 6 iterations to fully resolve remaining overlaps. The typical order of magnitude for the BBPGD iteration count is consistent with related studies [41], even though our model's collision-growth coupling introduces additional complexity.

The convergence behavior of a single representative ~~time step~~ timestep is illustrated in Figure 19b. The solver reaches the desired tolerance of $\varepsilon = 10^{-3}$ after approximately 280 iterations, with the residual directly corresponding to the maximum overlap between any two cells in the colony. The residual exhibits a highly oscillatory pattern, a well-known characteristic of BBPGD resulting from adaptive step-size selection [10, 30].

Since we use the ReLCP procedure [36], each simulation step may involve multiple BBPGD iterations until all overlaps are resolved. The overall energy evolution during this convergence process is shown in Figure 19c, illustrating that the solver steadily reduces the system's energy until the current overlaps are eliminated. Once this state is reached, new contacts are detected, the system's energy increases, and the process repeats. The system ultimately converges to a feasible, overlap-free configuration within six ReLCP iterations, requiring a total of 960 BBPGD steps. The plot also shows the total number of constraints resolved during the BBPGD iterations, which grows with each new ReLCP iteration. For a colony of size $R \approx 100$ with 36,244 cells, roughly 100,000 new constraints are generated per ReLCP iteration, reaching a maximum of about 600,000 constraints in the final phase.

There remains considerable potential to further improve the performance of BBPGD. As demonstrated in subsection 7.4, the step-size critically influences solver efficiency. We propose the following directions as open research problems not yet explored in the literature: more advanced adaptive schemes that directly leverage solver feedback, for instance by adjusting Δt based on the number of BBPGD iterations or the ReLCP convergence rate; and warm-start techniques that initialize the solver with multipliers from previous timesteps to reduce iteration counts.

7.6. Maximum Attainable Colony Size

In this section, we examine the largest colony sizes achievable within a 24-hour budget on 112 CPU cores of the `CoolMUC-4` cluster at $\lambda = 10^{-3}$. Only the hard model is considered, as the soft model exhibits severe overcrowding and overlap at these scales.

Figure A1 shows a representative snapshot at the maximum simulated size of $R = 260$ with 301,116 cells. The colony exhibits clear concentric density bands and a smooth boundary, indicating that the solver, contact handling, and domain decomposition remain stable at this scale.

The computational cost of reaching such colony sizes is summarized in Figure 20. The wall time grows rapidly with colony radius and closely follows a fifth-degree polynomial fit given by $T_{\text{wall}}\,[\text{h}] \approx 1.94 \cdot 10^{-11}\, R^5 + 0.37$. Since the number of particles scales approximately as $N \propto R^2$, this implies a total runtime scaling of $T_{\text{wall}}\,[\text{h}] \propto N^{2.5}$.

The solver workload is illustrated in Figure 21a, which displays the number of BBPGD iterations per timestep. At this scale, each timestep requires resolving about 5,131,551 contact constraints, distributed across 8 ReLCP phases. The number of BBPGD iterations grows roughly linearly with the number of particles, although two pronounced spikes of up to 25,000 iterations occur during the simulation. These spikes likely correspond to critical moments of large-scale rearrangements or near-jamming transitions, where the solver must resolve a particularly complex configuration of contacts.

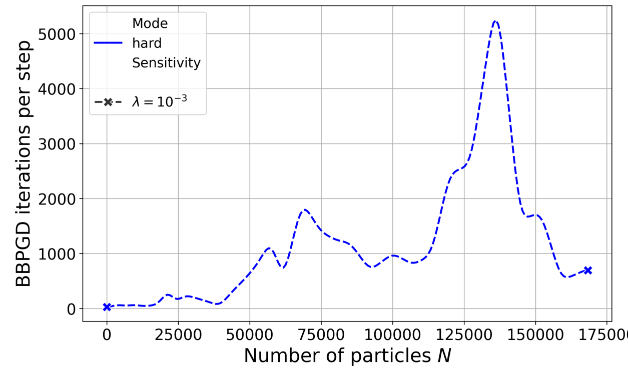
Figure 21b presents the radial distribution of the average cell length at $R = 260$. Clear wave-like patterns with a characteristic ring spacing of $\cancel{\xi \approx 25}$ $\xi \approx 25\,\ell_0$ are observed, corresponding to the ring spacing visible in Figure A1. This value is of the same order as the continuum prediction of Weady et al. [35], whose ring wavelength of $\approx 5\,\ell_0$ at $\lambda = 10^{-2}$ scales as $\lambda^{-1/2}$ and thus extrapolates to $\approx 16\,\ell_0$ at the $\lambda = 10^{-3}$ used here; the residual difference is consistent with our value being measured at finite colony size rather than in the asymptotic continuum limit.

Finally, Figure 21c shows the corresponding radial stress distribution. In contrast to the results in Figure 9, the simulated stress profile closely follows the analytical prediction of Weady et al. [36].

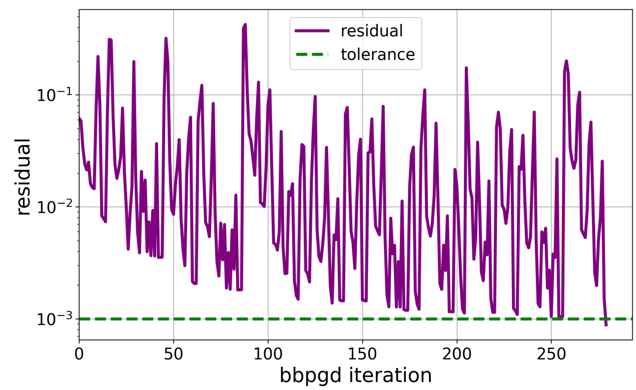
8. Discussion and Conclusion

This work presents a systematic comparison of hard (constraint-based) and soft (potential-based) collision models for simulating proliferating cell collectives within a unified computational framework, yielding key insights into the trade-offs between these two approaches.

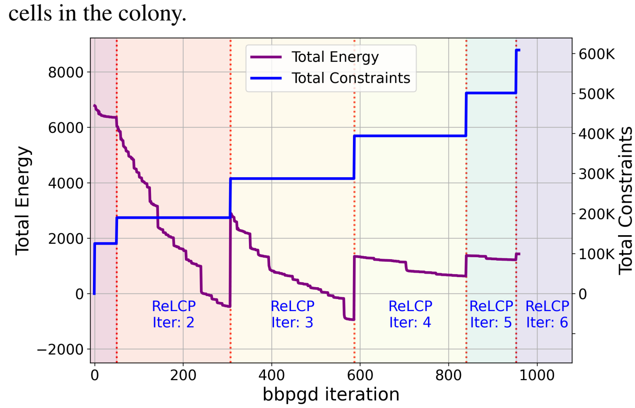
Both models successfully reproduce the concentric ring patterns and microdomain formation observed in experimental bacterial colonies at small scales, demonstrating that



(a) Number of BBPGD iterations per timestep as a function of particle count for a representative run at $R \approx 100$. Iterations remain roughly $\mathcal{O}(1000)$ on average, with occasional spikes up to $\mathcal{O}(5000)$.



(b) BBPGD residuals over iterations for a single timestep. The solver converges to tolerance $\varepsilon = 10^{-3}$ within approximately 280 iterations, showing highly oscillatory behavior. The residual directly represents the current maximum overlap between any two cells in the colony.



(c) System energy (Equation 16) throughout the ReLCP procedure during a single timestep. The energy consistently decreases as BBPGD iterations progress. After each set of constraints is resolved, new overlaps are detected, and the process repeats until a feasible configuration is reached within six ReLCP iterations. The rightmost segment (ReLCP Iter 6) shows the final feasible configuration where all constraints are resolved.

Figure 19: BBPGD convergence analysis for a typical timestep in a colony of size $R \approx 100$ with 36244 cells. (a) BBPGD iterations per timestep versus number of particles. (b) Residuals over iterations. (c) System energy over iterations.

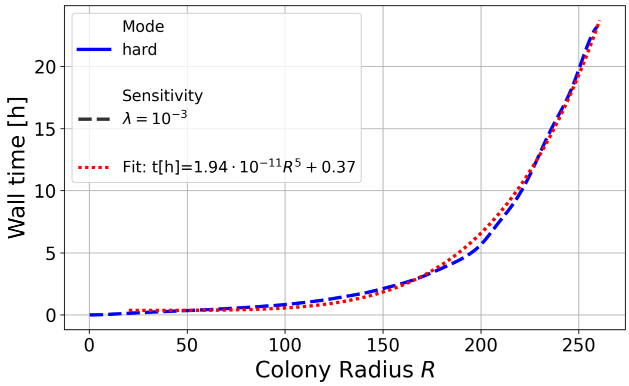
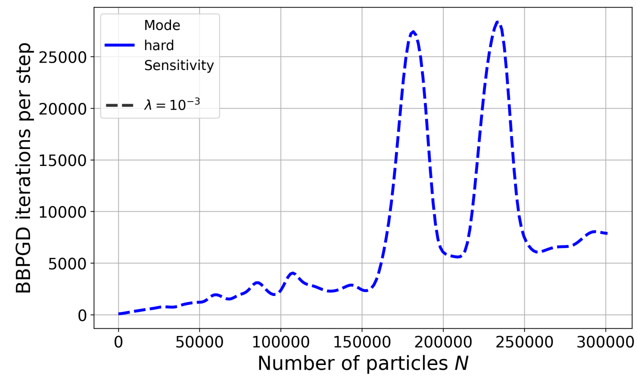


Figure 20: Wall time as a function of colony radius. The wall time roughly follows $T_{\text{wall}}\,[\text{h}] \approx 1.94 \cdot 10^{-11}\, R^5 + 0.37$.

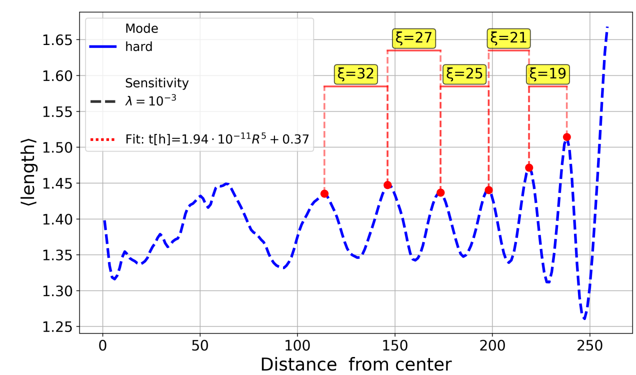
these phenomena emerge from growth-mechanics feedback and are robust to the specific details of collision resolution. This robustness justifies the use of computationally simpler soft models for studies focused on colony-level pattern formation at small radii.

At the microscopic level, however, the models differ significantly. The hard model enforces strict non-overlap, maintaining a packing fraction of about 0.9 and producing reliable stress distributions and microdomain structures. In contrast, the soft model allows significant, unphysical overlap, with packing fractions exceeding 5 in colony centers. This leads to distorted microdomain morphology and elongated, unrealistic cell bundles, particularly in large colonies or under low stress-sensitivity conditions. These artifacts limit the soft model's applicability for detailed analyses of cell-scale organization or mechanical stress distributions. ~~Reducing the step-size further, based on cell overlap, may alleviate this issue, but would come with reduced computational performance~~Because this overlap reflects the finite stiffness of the repulsive potential rather than temporal under-resolution, it cannot be removed by smaller step-sizes alone; a stiffer potential would reduce it, but only at a further, prohibitive computational cost.

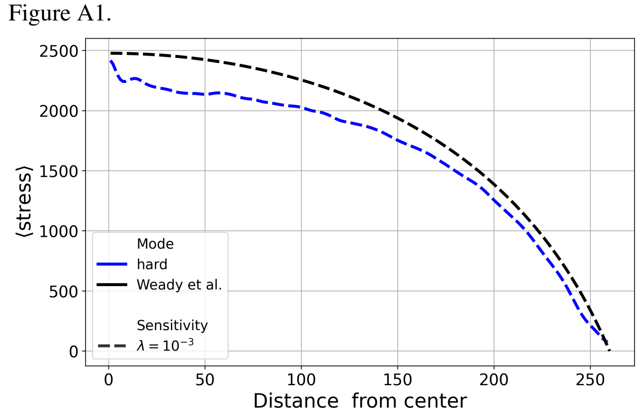
The choice between the two paradigms is ultimately dictated not only by computational efficiency but also by the physics one wishes to capture. Soft potential-based models represent cells as elastically deformable bodies and resolve force-based mechanical interactions directly, which makes them a natural choice when cell deformability or compliance is itself the object of study, as is done in several works [4, 13, 14, 16]. The hard model instead treats cells as rigid; a controlled degree of deformation could nonetheless be reintroduced by relaxing the overlap tolerance ε, which would additionally reduce the number of constraint-solver iterations and thereby further improve the hard model's efficiency. A rigorous physical comparison of such tolerance-controlled deformation against an explicit soft potential is left for future work. On the other hand, by increasing the contact stiffness k_{cc}, a soft model can be used to model



(a) BBPGD iterations per timestep as a function of particle count for the hard model. Each timestep at $R = 260$ resolves about 5,131,551 constraints over 8 ReLCP phases, requiring on average 7,800 iterations, with occasional peaks up to 25,000.



(b) Radial distribution of the average cell length for the hard model at $R = 260$. The oscillatory pattern has a characteristic ring spacing of $\xi \approx 25$, consistent with the radial ring spacing visible in Figure A1.



(c) Radial stress profile of the hard model at $R = 260$. The simulated stress closely follows the analytical prediction from [36], confirming global mechanical balance across the colony.

Figure 21: Analysis of maximum colony size for the hard model under a fixed 24-hour computational budget. (a) Solver iterations as a function of particle count. (b) Radial oscillations in cell length and ring spacing. (c) Radial stress distribution compared to analytical predictions.

mostly-rigid cells with realistic packing fractions, but at a far greater computational cost.

Even without this, performance benchmarking shows that the hard model consistently outperforms the soft model across all colony sizes, achieving up to 9.36× faster runtimes on 112 cores. This advantage arises from its ability to use step-sizes roughly 30 times larger ($\Delta t_{\text{hard}} \approx 3 \cdot 10^{-4}$ vs. $\Delta t_{\text{soft}} \approx 10^{-5}$), which amortizes communication costs over more computational work. The hard model also exhibits better parallel scaling, maintaining reasonable speedup up to 64 cores, whereas the soft model scales poorly beyond 32 cores. Within a fixed 24-hour computational budget on 112 cores, the hard model simulated colonies up to $R \approx 260$ with approximately 301,116 cells while preserving physical accuracy.

This work only considered two-dimensional simulations, and more work is needed to compare the performance of the two models for three-dimensional simulations. We expect the hard model's efficiency advantage to persist, and potentially widen, in three dimensions: Brendel et al. [7] suggest that the computational cost of the hard model relative to that for the soft model is proportional to $N^{\frac{2}{\mathbb{D}}}$, where $\mathbb{D}$ is the dimension. This is, however, a simplification of the relationship we would expect, assuming non-active spherical particles and not considering the implications upon parallelization. Nonetheless, we expect still that the soft model's stiffness-limited step-size would remain the dominant bottleneck, whereas the hard model's larger stable steps continue to amortize this cost. The main additional challenge lies not in contact detection, which is already three-dimensional, but in the scalability of the constraint solver and the domain decomposition under the increased contact density of packed 3D aggregates. Whilst the domain decomposition is two-dimensional, Warren et al.'s three-dimensional colony simulation results in a height ranging between half to a fifth the radius [34], and so the two-dimensional decomposition may still be suitable.

The use of an adaptive timestepping algorithm based on the Courant-Friedrichs-Lewy condition proved essential for numerical stability, dynamically selecting step-sizes suited to the colony state. For the hard model, an optimal CFL factor of approximately 2 was identified, balancing solver iterations per step against the total number of steps and minimizing runtime. For the soft model, we would expect smaller step-sizes, whether through criteria based on the maximum velocity or the current overlap, to reduce overlap. However, this would come at an increased computational cost, when the soft model is already shown to be computationally more expensive in the scenarios investigated.

The only related work for the soft model that provided step-size information, Warren et al. [34], gave a fixed step-size of 5×10^{-5} – higher than that given by our adaptive timestepping algorithm for the later time periods of the simulation, which experienced the most overlap – which may suggest that a nutrient-driven growth constraint enables low-overlap simulations with the soft-model at higher step-sizes, but further work is needed to verify this.

In summary, within the regimes examined here – two-dimensional, mechanically-driven colonies of a single rod-shaped cell type without nutrient or biochemical feedback, and a single soft-contact implementation – the hard model is ~~strongly preferred for nearly all~~ preferred for most applications, as it enforces realistic packing densities, yields accurate stress distributions, and provides superior computational performance. The soft model may still be useful for exploratory studies of very small colonies ($R \lesssim 50$) at high stress-sensitivity, where packing artifacts are limited and only pattern formation is of interest. However, its severe physical limitations, including unphysical overcrowding, distorted stress distributions, and artificial cell bundles, make it unsuitable for analysis or large-scale simulations.

9. Future Work

The computational challenges identified in this work suggest several promising research directions to enhance the scale and efficiency of simulating proliferating cell collectives.

9.1. Improved Parallel Scalability

Communication overhead at high core counts limits scalability for both hard and soft models. Future work should focus on improving communication patterns and domain decomposition to reduce MPI bottlenecks and better balance workloads across ranks. Strategies could include asynchronous communication, overlapping computation with communication, and hierarchical or adaptive decomposition schemes that account for particle density and colony growth. Hybrid MPI+OpenMP parallelism could further reduce communication costs by allowing multiple threads per rank to share memory. PETSc supports limited hybrid MPI+OpenMP parallelism via its thread-based backends; leveraging this more fully, or adopting complementary libraries for the thread-parallel portions, could reduce per-rank memory footprint and communication overhead.

9.2. GPU Acceleration via PETSc

For the hard model, the primary computational bottleneck at large scales is the BBPGD solver. Building on the work of [32], future work could leverage PETSc's built-in GPU support to offload linear algebra operations to GPUs with minimal code changes. This approach has been shown to provide significant performance improvements in large-scale simulations where the constraint solver dominates runtime.

9.3. Model Extensions

Future work could also investigate extensions to the current models to capture additional biological phenomena. For instance, external boundaries, nutrient fields, or chemotactic signaling could be incorporated to study their effects on colony growth and pattern formation. Most of these extensions can be integrated into the existing framework with minimal changes to the collision models [40]. Additionally, future work could further examine the influence of the stress-sensitivity parameter λ on the emergence, size, and stability of microdomains in the hard model, following an approach similar to the analyses presented by You et al. [43]. Furthermore, many soft model related works limit cell growth through a depleting nutrient model. It may be that this limits cell overlap in the soft model, potentially improving that model relative to the hard one.

10. Supplementary Materials

All high-resolution figures and supplementary videos are available online at `https://doi.org/10.5281/zenodo.19931308`.

11. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used generative AI tools, including ChatGPT (OpenAI)~~and~~, GitHub Copilot (GitHub/Microsoft), and Claude (Anthropic), in order to improve spelling, grammar, and phrasing during manuscript writing and to assist with software development. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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CRediT authorship contribution statement

Manuel Lerchner: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing — original draft, Writing — review and editing. **Samuel James Newcome:** Conceptualization, Supervision, Writing -— review and editing.

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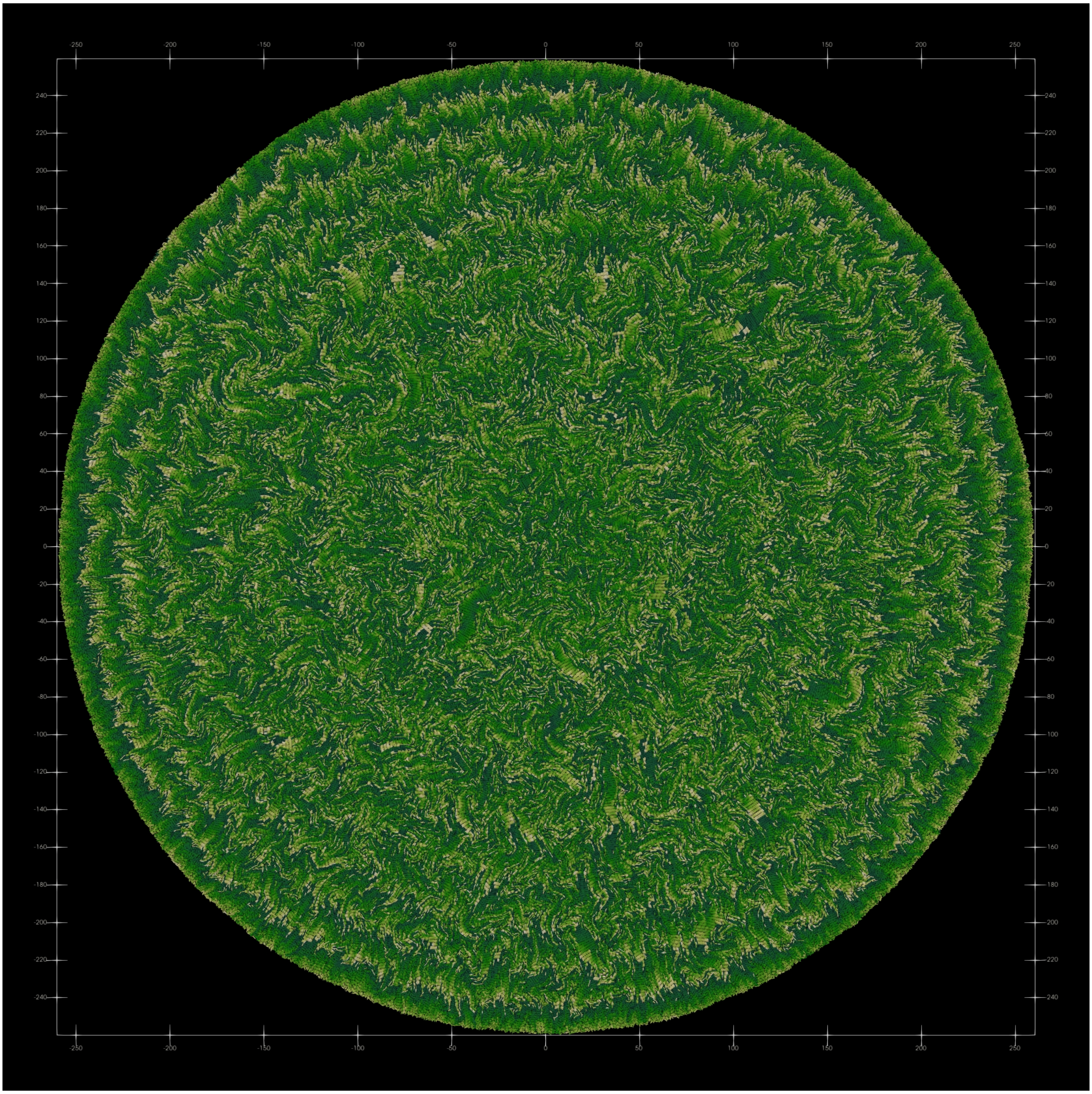

Figure A1: Snapshot of largest attainable colony size using the hard collision model with $\lambda = 10^{-3}$ after 24h of wall time on 112 ranks on the CoolMUC-4 cluster. The colony has a radius of approximately $R \approx 260$ and consists of $301{,}116$ cells. The color indicates the length of the cells (short cells are darker, longer cells are lighter). Clear concentric ring patterns are visible, similar to those observed in smaller colonies (See Figure 4).